

The weight of cognitive abilities: Testing and re-testing strategic games in adolescents*

Pablo Lomas[†]

April 27, 2026

Abstract

This paper examines how strategic decision-making in children and adolescents depends on cognitive abilities. To this end, we implemented two one-shot economic games (one static and one dynamic) in simplified two-player versions. The static one, *CARDS* ($N = 2,718$), is adapted from the Beauty Contest Game, while the dynamic one, *PIGGY* ($N = 653$), used a task based in the Chicken Game with an accumulation process to test intertemporal strategic decision-making. As outcome variables, we constructed measures of Nash Equilibrium play, trembling, and rationality accuracy, defined as the distance from playing the Nash Equilibrium. Cognitive abilities were measured using a composite indicator based on three tasks: a Cognitive Reflection Test (CRT), a test of mathematical and financial skills, and a set of questions assessing knowledge of probabilities. Participants were secondary school students in Spain aged 10 to 17, and they also completed tasks eliciting time-discounting preferences and risk attitudes. We report four main findings. First, subjects play the Nash Equilibrium more frequently in *CARDS* than in *PIGGY*. Second, in both designs, cognitive abilities significantly predict strategic decision-making. Third, in *PIGGY*, strategic behavior is better predicted by cognitive abilities than in *CARDS*. Fourth, we confirm prior literature showing no age effects on optimal strategic decision-making among children and adolescents.

Keywords: Strategical thinking, Adolescents, Cognitive abilities, Nash equilibrium.

JEL codes: C7, C93, D90, D91

*I am grateful to Laura Giraldo, Emilio Nieto, Paula Piña and Luis Rivero for their invaluable assistance with data collection. I also thank Pablo Brañas-Garza, Diego Andrés Jorrat, and Natalia Jiménez Jiménez for their helpful comments and insightful feedback, which contributed to improving earlier stages of this research. In addition, I sincerely appreciate the collaboration of all participating schools for their continued support and engagement throughout the project. This research was supported by the Spanish Ministry of Economy and Competitiveness (PID2021-126892NB-I00), Excelencia-Junta (PY-18-FR-0007), and the Agencia Andaluza de Cooperación Internacional para el Desarrollo (AACID-0I008/2020). The experimental procedures were approved by the Ethics Committee of Loyola University Andalusia.

[†]Loyola University Andalusia. Corresponding author. Email: pablodelomas1997@gmail.com

1 Introduction

Understanding how individuals make choices under uncertainty, evaluate trade-offs, and anticipate others' behaviour has been a central concern across economics, psychology, and neuroscience. Research in behavioural economics and cognitive psychology has shown that decision-making is not purely rational but systematically shaped by heuristics, emotions, and contextual framing (kahneman2011fast; Camerer, 2003; Thaler, 1980).

To take advantage of interactions within the environment (especially in the human interaction) we use the ability to anticipate others. For example, investors base their decisions on their beliefs about the return expectations of other investors (Egan et al., 2014; Rangvid et al., 2013), or the capacity to success within a firm depends on the ability of players to anticipate the actions of their peers (Brandts & Cooper, 2006).

Building on this general understanding of decision-making, adolescence represents a particularly insightful stage to study how these processes emerge and consolidate. During this period, individuals transition from intuitive and affect-driven responses towards more reflective and goal-oriented reasoning (Crone & Dahl, 2012; Steinberg, 2005). This developmental shift provides a natural laboratory to observe how cognitive control, foresight, and social reasoning interact under varying incentives. Investigating these processes experimentally allows researchers to uncover how cognitive and socio-emotional mechanisms jointly influence decision-making (Brocas & Carrillo, 2020a; Sutter et al., 2019).

With a few exceptions (Brañas-Garza et al., 2012, 2026a; Brocas & Carrillo, 2018, 2020a, 2020c, 2021b; Czermak et al., 2016; Fe et al., 2022; Glätzle-Rützler et al., 2025; Sher et al., 2014), there is limited research on how children and adolescents adopt others' perspectives and how this translates into their own strategic decision-making (Hermes & Schunk, 2022). Nevertheless, previous studies have examined children's behavior in simpler interactive games such as dictator, ultimatum, trust, and public goods games (Harbaugh & Krause, 2000; Harbaugh et al., 2003; Murnighan & Saxon, 1998).

But a growing body of literature has explored the developmental trajectory of strategic reasoning. Brocas and Carrillo (2018, 2020b, 2021b) show that iterative thinking and belief formation emerge gradually from childhood to adolescence, paralleling the maturation of executive functions and perspective-taking abilities (Sher et al., 2014). In addition, children aged 9-11 years have been shown to be capable of detecting *faux pas* (i.e. situations where one person unintentionally hurts another emotionally due to lack of information), and this belongs to the development of *Theory of Mind* (*ToM*, henceforth) (Baron-Cohen et al., 1999; Stone et al., 1998).

Building on this literature, strategic behavior can be understood as the ability to anticipate others' actions and adjust one's choices accordingly. In experimental game-theoretic settings, such ability is commonly measured by equilibrium play, particularly the proportion of participants who play the Nash equilibrium (*NE*,

henceforth). From this perspective, reaching the *NE* provides an observable benchmark for optimal strategic reasoning.

Consistent with this view, Fe et al. (2022) and Crawford et al. (2013) show that coordination and equilibrium play critically depend on cognitive ability, working memory, and numeracy. Brañas-Garza et al. (2012) find that cognitive reflection predicts higher levels of strategic sophistication in adults, while Fe et al. (2022) report that these skills continue to develop throughout adolescence. More broadly, understanding how cognitive abilities interact with *ToM* remains an active area of research (Birch et al., 2017; Carlson & Moses, 2001; Carlson et al., 2004; Davis & Pratt, 1995; Flynn, 2007; Hala et al., 2003; Hughes, 1998; Landry et al., 2002; Moses & Carlson, 2004; Müller et al., 2012; Russell, 2013; Zelazo et al., 1997).

To further refine the measurement of strategic behaviour and to better understand how anticipatory skills relate to cognitive reflection and other developmental variables, such as cognitive abilities, it is essential to design experimental methods that effectively elicit optimal decision-making. In other words, experimental tasks must provide participants with clear and meaningful cues that reveal their underlying reasoning processes when facing strategic or dynamic environments. These abilities are linked to many benefits, such as better emotional well-being, academic achievement, and stronger relationships (see Repacholi and Slaughter (2004) for a summary). Within this framework, we address three main research questions: (i) how strategic behaviour relates to cognitive abilities, (ii) whether this relationship differs across experimental designs, and (iii) whether it varies with age.

To this end, we conducted two one-shot experiments. The first experimental design (*CARDS*) is a simplified version of the *Beauty Contest Game* (henceforth, *BCG*), originally introduced by Nagel, 1995¹. In this task, participants choose one card among five, with payoffs depending on the counterpart’s choice. While previous research shows that children can understand the logic of the *BCG* and that best-response behaviour is linked to cognitive ability, there is still limited evidence on the extent to which children and adolescents anticipate others’ decisions and display strategic reasoning (Hermes & Schunk, 2022). The second experimental design (*PIGGY*) examines intertemporal strategic decision-making through a dynamic accumulation setting related to the Chicken Game (Rapoport & Chammah, 1966) and the War of Attrition framework (Fudenberg & Tirole, 1991; Smith, 1974). In this task, participants decide when to stop a growing virtual piggy bank within 30 seconds: acting after the opponent yields the full accumulated amount, whereas waiting until the deadline leads to an equal split, capturing the trade-off between waiting and acting under strategic uncertainty (Friedman & Oprea, 2012; Oprea et al., 2011).

Together, *CARDS* and *PIGGY* provide a multidimensional assessment of adolescents’ ability to anticipate others’ intentions in strategic settings. Such anticipation—linked to higher-order reasoning²—requires the use of *ToM* to maximize

¹The original game consisted on N decision-makers simultaneously choose a number between 0 and 100 and the participant with the number closest to $p * \bar{x}$ wins a fixed prize.

²“*She thinks that I think that, then I think that she thinks that I think that, and she thinks that I think that she thinks that I think that,*” and so on.

expected payoffs. Both designs follow best practices in developmental experimental economics (Alfonso et al., 2023a; Brocas & Carrillo, 2020a), incorporating simplified instructions, visual aids, and reduced choice complexity to ensure clarity and minimize cognitive load.

Importantly, all outcomes in the experiment were hypothetical, and participants were clearly informed that there would be no real monetary payments. While a substantial body of evidence suggests that incentive type does not significantly alter elicited preferences in adolescent samples (Alfonso et al., 2025; Brañas-Garza et al., 2021, 2026b), we acknowledge that a parallel strand of the literature documents meaningful differences between real and hypothetical payoffs. Incentive effects tend to be strongest in tasks involving risk elicitation and effort provision, where real stakes are found to produce more calibrated behavioral responses (Camerer & Hogarth, 1999; Holt & Laury, 2002; Xu et al., 2018). Evidence also suggests that adolescents may be comparatively more sensitive to real monetary incentives in these domains (Kray et al., 2018). Crucially, however, these effects appear to be domain-specific: strategic coordination tasks consistently show minimal behavioral differences between incentivized and hypothetical conditions (Brañas-Garza et al., 2026b).

To properly analyze the influence of cognitive abilities on strategic thinking, we constructed a composite index based on three cognitive skill measures. These included the Cognitive Reflection Test (*CRT*)³, a financial literacy test (*FinAb*)⁴ and the inclusion of three violations within a probabilistic reasoning task (*Del*)⁵.

Frederick (2005) showed that individuals who perform better on *CRT* tend to be more patient and display a greater willingness to take risks in gain domains. Oechssler et al. (2009) found that subjects with lower scores on cognitive tests are more likely to commit the conjunction fallacy and to exhibit greater conservatism when updating probabilities. In this regard, cognitive reflection can be understood as a combination of cognitive ability and the disposition to engage in deliberate decision-making and judgment (Finucane & Gullion, 2010). Accordingly, Toplak et al. (2011) showed that *CRT* captures important aspects of rational behavior that are not measured by other intelligence tests.

For this study, we not only analyze whether participants play the *NE*, but also the extent to which they approximate it. Since prior literature shows that individuals rarely play the *NE* exactly (Baddeley, 1966), we incorporate two additional measures capturing deviations from optimal play: *Trembling* and *Rationality*. *Trembling* accounts for small, potentially accidental deviations from the best response, following the idea of unintended mistakes (Selten, 1983), while *Rationality* captures how closely participants’ decisions align with the *NE*, providing a measure of their overall level of strategic reasoning.

This study finds four main results. First, subjects play the *NE* more frequently

³Originally developed by Frederick (2005) and by Thomson and Oppenheimer (2016).

⁴Composed of three questions from Alfonso et al. (2025).

⁵Adapted from Delavande and Rohwedder (2008).

in the static game based on the *BCG (CARDS)* than in the dynamic based on the *Chicken Game (PIGGY)*. Second, cognitive abilities are a key determinant of strategic decision-making in both experimental designs. Third, these cognitive abilities are more sensitive predictors in *PIGGY* than in *CARDS*. Fourth, we find no evidence that age predicts strategic decision-making, and we do not observe a developmental progression in strategic thinking.

In this regard, we contribute to the literature on strategic decision-making in children and adolescents in three ways: (i) by designing a tool to study optimal decisions among children and adolescents, (ii) by analyzing the role of cognitive abilities in strategic decision-making, and (iii) demonstrating that cognitive abilities has an essential role for playing *NE* in simultaneous time-limited strategic games.

By situating these results within the broader literature on adolescent decision-making, this study contributes to understanding how reasoning, learning, and strategic adaptation develop over time. It also provides methodological insights on how to design and refine experiments for younger populations, ensuring that observed behaviours accurately reflect underlying cognitive processes. Ultimately, this work bridges developmental and experimental economics, illustrating that adolescent decision-making is neither random nor irrational, but a structured reflection of growing cognitive capacity within appropriately designed experimental environments.

The rest of the paper is organized as follows. Next section describes the experiment, the sample experimental designs, and metrics. Third section shows the results and fourth concludes.

2 The experiment

In the last decades, preferences and skills in children and adolescents has been analyzed and documented in experiments (see for a summary Kautz et al. (2014), for the reason that a comprehensive knowledge of the process of decision making in children determines adult life outcomes (Sutter et al., 2019).

The overarching aim of this study is to examine how adolescents' cognitive skills, measured through cognitive reflection and numeracy tests, relate to performance across two one-shot games. The study of cognitive skills related to behavior is particularly important, as strong correlations have been documented between cognitive abilities and wages, educational attainment, employment, and occupational status (Cawley et al., 2001), as well as with smoking, illegal drug use, and teenage pregnancy (Heckman et al., 2006). On the other hand, our decision to use one-shot experiments stems from the fact that we do not aim to observe learning within the game, but rather to capture spontaneous decisions in a population that has not yet reached full neural maturity. Increasing the number of rounds, and thus experience with the game, may have a fundamental impact on decision-making (Ponti, 2000). Some studies of the Centipede Game employ a single round (Bornstein et al., 2004; Levitt et al., 2011; Palacios-Huerta & Volij, 2009), while others use multiple rounds. In fact, many repeated-round experiments reveal small learning effects (Fey et al., 1996; McKelvey & Palfrey, 1992; Pulford et al., 2017).

This study was conducted as part of a broader research initiative aimed at understanding how children and adolescents make decisions in economic and social environments (Alfonso et al., 2025). All tasks were implemented in classroom settings during regular school hours, under a standardized protocol applied in secondary schools in Spain. The experiment was fully integrated into the schools’ academic curriculum. In accordance with the ethical guidelines established by the Institutional Review Board (IRB), an opt-out consent procedure was used. All participants completed the tasks while being supervised by an adult (typically a teacher or trained facilitator).

A total of 13 schools and 162 classes from Spain participated in the study. Participants were students aged 10 to 17 years, enrolled across middle school, and the structure of the tasks was specifically adapted to this age group. Sessions were held between October 2021 and December 2022. In line with the recommendations provided by Brocas and Carrillo (2020b) for experimental research with minors, the design was simplified to accommodate limited attention spans and all tasks were presented in a clear, intuitive and visually engaging format.

The experimental interface was delivered through SAND (Social Analysis and Network Data), a web-based platform developed by Kampal Data Solutions for administering behavioral experiments in field settings. SAND is optimized for adolescents and ensures usability across devices, along with strict data protection and anonymity protocols. Participants completed the tasks individually and without interacting with peers. Most participants completed the tasks using their own smartphones (59%), followed by school-provided desktop computers (15.1%) and school tablets (25.9%), depending on the technological resources available at each educational center.

Before starting, participants received a collective briefing, after which they completed a series of decision-making tasks. These tasks covered a wide range of domains, including temporal discounting, economic elicitation procedures, social network structures, basic knowledge of mathematics and probability, and several psychological measures related to emotions and individual preferences. Importantly, all outcomes in the experiment were hypothetical, and participants were clearly informed that there would be no real monetary payments. While a substantial body of evidence suggests that incentive type does not significantly alter elicited preferences in adolescent samples (Alfonso et al., 2025; Brañas-Garza et al., 2021, 2026b), we acknowledge that a parallel strand of the literature documents meaningful differences between real and hypothetical payoffs, particularly in tasks involving risk elicitation and effort provision (Camerer & Hogarth, 1999; Drobner et al., 2024; Holt & Laury, 2002; Xu et al., 2018). Importantly, adolescents may be comparatively more sensitive to real monetary incentives in these domains (Kray et al., 2018). These effects, however, appear to be domain-specific: strategic and social preference tasks—most similar in nature to those employed in the present study—tend to show minimal qualitative differences between incentivized and hypothetical conditions, even when quantitative magnitudes may differ (Brañas-Garza et al., 2026b; Masclet & Rebière, 2021).

2.1 Sample

The sample comprises a mix of public and semi-private schools, ensuring geographic and socioeconomic diversity across the dataset. In total, we find 9 public and 4 semi-private schools. The sample also contains students enrolled in 1st to 4th years of compulsory secondary education in Spain, and under the age of 18⁶.

Each participant completed only *CARDS* or *PIGGY*, and no individual was invited to participate in more than one experimental task, resulting this research in a study between subjects. The assignment to each task was not randomized, and it was implemented at the school level. Schools 1–10 were assigned to *CARDS* and schools 11–13 to *PIGGY*. This allocation was driven by practical and organizational considerations during field implementation, including coordination with school schedules and the sequential rollout of experimental sessions. Although additional schools were initially planned to be included under the *PIGGY* design, data collection was constrained by the end of the academic year, which limited the final number of participating schools in this condition. As a result, each observation in the data set corresponds to a unique participant exposed to a single experimental design only, and the resulting data can be interpreted as a between-subject design with two non-overlapping groups. See Table 1 for descriptive statistics.

Gender distribution was approximately balanced (see Table 2), and participation rates were nearly universal within classrooms, as no student opted out under the consent procedure. A total of 3,554 subjects participated in the experiment, with the largest subsample assigned to *CARDS* ($n = 2,718$), followed by *PIGGY* ($n = 653$). The difference in sample sizes reflects the school-level assignment and the timing constraints affecting the implementation of the *PIGGY* sessions. We find an overall female representation of 47.67% in all sample. The proportion of women was similar under both tasks: 47.42% in *CARDS* and 48.70% in *PIGGY*. To assess whether gender composition was balanced within each experimental condition, we conducted one-sample proportion tests comparing the share of female participants to 50%. In *CARDS*, the proportion of female participants is slightly below 50% ($\chi^2(1) = 7.11$, $p = 0.0077$), although the magnitude of the difference is small (47.4% female). Results also indicate that the proportion of female participants in *PIGGY* does not differ from 50% ($\chi^2(1) = 0.39$, $p = 0.531$).

We also tested whether the distribution of gender varied by academic grade within each experimental design to ensure that gender balance was preserved across age cohorts. Pearson chi-square tests of independence show no statistically significant differences in gender composition across grades in either sample (*CARDS*: $\chi^2(3) = 3.08$, $p = 0.380$; *PIGGY*: $\chi^2(3) = 2.79$, $p = 0.426$).

⁶In Spain, students over 18 years of age are not legally permitted to remain in standard secondary education programs and must instead enroll in adult education pathways.

Table 1: Descriptive statistics

Variable	Mean	SD	Range	N
<i>CARDS</i>	0.8063	0.3953	0 - 1	2718
<i>PIGGY</i>	0.1937	0.3953	0 - 1	653
<i>Age</i> *	12.8321	1.4295	10 - 17	3371
<i>Female</i>	0.4767	0.4995	0 - 1	3371
<i>7th grade</i> [†]	0.2880	0.4529	1 - 4	3371
<i>8th grade</i>	0.3044	0.4602	1 - 4	3371
<i>9th grade</i>	0.2026	0.4020	1 - 4	3371
<i>10th grade</i>	0.2050	0.4037	1 - 4	3371
<i>Class size</i>	29.0982	2.5666	17 - 34	3371
<i>School 1</i>	0.0724	0.2592	1 - 13	3371
<i>School 2</i>	0.1578	0.3646	1 - 13	3371
<i>School 3</i>	0.0492	0.2164	1 - 13	3371
<i>School 4</i>	0.0344	0.1823	1 - 13	3371
<i>School 5</i>	0.0679	0.2517	1 - 13	3371
<i>School 6</i>	0.1095	0.3123	1 - 13	3371
<i>School 7</i>	0.0249	0.1559	1 - 13	3371
<i>School 8</i>	0.0670	0.2501	1 - 13	3371
<i>School 9</i>	0.1012	0.3016	1 - 13	3371
<i>School 10</i>	0.1219	0.3272	1 - 13	3371
<i>School 11</i>	0.0276	0.1638	1 - 13	3371
<i>School 12</i>	0.0365	0.1875	1 - 13	3371
<i>School 13</i>	0.1296	0.3360	1 - 13	3371
<i>Public school</i> [§]	0.7188	0.4497	1 - 2	3371
<i>Semiprivate school</i>	0.2812	0.4497	1 - 2	3371
<i>Computer</i>	0.1508	0.3579	0 - 3	3368
<i>Phone</i>	0.5900	0.4919	0 - 3	3368
<i>Tablet</i>	0.2586	0.4379	0 - 3	3368
<i>CRT</i> [¶]	1.4994	0.8024	0 - 3	3364
<i>FinAb</i>	1.0648	0.8345	0 - 3	3348
<i>Del</i> ^α	2.1851	0.8596	0 - 3	3317
<i>CogAb</i> ^β	4.7521	1.7472	0 - 9	3291
<i>Risky</i> ^γ	3.4978	1.0104	0 - 6	3369
<i>Patience</i> ^ω	3.2637	2.0445	0 - 6	3368

* Measured in years.

[†] 7th to 10th Grade correspond to the four levels of secondary compulsory education in Spain, where it is not possible to be above 17 years of age.

[§] There is no private school in the sample.

[¶] *CRT* follows the three questions task introduced by Frederick (2005) and adapted by Thomson and Oppenheimer (2016). The items used are shown in Figure 5 in Supplementary Materials.

^{||} *FinAb* is composed of three questions regarding basic operations and interest rates for measuring financial abilities from Alfonso et al. (2025). The items used are shown in Figure 6 in Supplementary Materials.

^α *Del* sums a total of three no-violations from a probabilistic reasoning task adapted from Delavande and Rohwedder (2008). Each no-violation adds 1 point to the index. The items used are shown in Figure 10 in Supplementary Materials.

^β *CogAb* is a composite index that includes the sum of *CRT*, *FinAb* and *Del*.

^γ To elicit risk preferences, we include *Risky*: the number of risk options in the graphic *The Gumball Machine task*'s version of Holt and Laury (2002) introduced by Vasco and De Francisco (2023). The items used are shown in Figure 18 in Supplementary Materials.

^ω To elicit time preferences, we used the visual version of *The Truck Task*: a Multiple Price List of Collier and Williams (1999) modified by Alfonso et al. (2023a). *Patience* is defined as the number of patient choices. The items used are shown in Figure 11 in Supplementary Materials.

Table 2: Number of participants by experimental design

Task	Male	Female	Other	% Female	Total
<i>CARDS</i>	1,310	1,289	119	47.42	2,718
<i>PIGGY</i>	321	318	14	48.70	653
<i>Total</i>	1,631	1,607	133	47.67	3,371

In the next section, we present a detailed description of the experimental tasks and provide additional tables detailing the sample composition by academic grade and gender.

2.2 Experimental designs

2.2.1 *CARDS*

Participants played one-shot strategic game with an anonymous partner. They were randomly matched in pairs and simultaneously chose a card $i \in \{1, 2, 3, 4, 5\}$, without communication. The cards were always presented in the same fixed order, from left to right. They were told that their counterpart would simultaneously view the same set of cards and make their own selection. Importantly, all payoffs were hypothetical and no real monetary incentives were provided based on participants' choices. A full account of the game's structure and the wording used can be found in Box 1 in the Appendix .

Each participant was assigned the role of *Player A*, while the counterpart was assigned the role of *Player B*. Importantly, all experimental subjects always played in the role of *Player A*, while *Player B* represents the matched counterpart.

Let i and j denote the choices of *Players A* and *B*, respectively. Payoffs were determined according to the following rules:

- If both players selected the same card number ($i = j$), *Player A* received $\pi_A(i, j) = i$ points, while *Player B* received $\pi_B(i, j) = 0$.
- If players selected different numbers ($i \neq j$), each player received the value of her own choice, that is, $\pi_A(i, j) = i$ and $\pi_B(i, j) = j$.

See in Table 3 the payoff matrix for *CARDS*.

Table 3: *CARDS* payoff matrix

A / B	1	2	3	4	5
1	(1 , 0)	(1 , 2)	(1 , 3)	(1 , 4)	(1 , 5)
2	(2 , 1)	(2 , 0)	(2 , 3)	(2 , 4)	(2 , 5)
3	(3 , 1)	(3 , 2)	(3 , 0)	(3 , 4)	(3 , 5)
4	(4 , 1)	(4 , 2)	(4 , 3)	(4 , 0)	(4 , 5)
5	(5 , 1)	(5 , 2)	(5 , 3)	(5 , 4)	(5 , 0)

Note: Red color represents best responses for Player A while blue color represents best responses for Player B.

This task is an adaptation of the "*Beauty Contest Game*" (henceforth, *BCG*), in which players must anticipate the reasoning levels of others in order to optimize their own strategy. In the classic version of the *BCG* (Nagel, 1995), players are asked to choose a number between 0 and 100, aiming to select a number closest to a fraction (commonly $2/3$) of the average of all players' choices. Optimal play in such games requires iterated reasoning, thinking about what others will think others will do, and so on, showing their *ToM* capacity.

In the current implementation, strategic reasoning is similarly required: participants must consider which card number a rational partner is likely to choose in order to avoid coordination (i.e. choosing the same card as the counterpart) and maximize their own payoff. As in the classic *BCG*, success depends on predicting the level of reasoning used by others. Recent studies showed that children can understand the dynamism of *BCG* and how their ability to play best response is correlated with their cognitive ability, but there is little about in a *BCG* to what extent children and adolescents are able to anticipate others' decisions and measure their capacity in strategic decision-making (Hermes & Schunk, 2022).

Table 4 presents the distribution of participants assigned to *CARDS*, disaggregated by gender and academic grade.

Table 4: Number of participants in *CARDS* by gender and school grade

Gender / School year	7th	8th	9th	10th	Total
Age assumed	11-12	12-13	14-15	16-17	
Male	387	393	263	267	1,310
Female	400	408	246	235	1,289
Other	29	45	18	27	119
% Female	49.02	48.23	46.68	44.42	47.42
Total	816	846	527	529	2,718

Note: These four academic years correspond to compulsory secondary education in Spain (*Educación Secundaria Obligatoria, ESO*), which students over 18 years of age cannot enroll in.

2.2.2 PIGGY

As in *CARDS*, participants were randomly matched in pairs, but this time they interacted in a continuous-time one-shot waiting game. Each participant was informed that they would be anonymously paired with another participant. In each pair, subjects were assigned the role of Player A, while the matched counterpart was assigned the role of Player B. Importantly, as also in *CARDS*, all experimental subjects always played in the role of Player A, whereas Player B represents the matched participant in the interaction. As in *CARDS*, all payoffs were hypothetical. Participants were informed of the rules and incentives, but no real monetary compensation was provided based on their performance. Box 2 in the Appendix provides the full description of the game and the exact instructions shown to participants.

Each player chose when to press a button in order to stop the accumulation of money in a virtual piggy bank, for a total time of 30 seconds. Let $t_i \in \{0, 1, \dots, 30, \emptyset\}$

denote the second at which Player $i \in \{A, B\}$ pressed the button, where $t_i = \emptyset$ represents that the player did not press the button within the 30-second limit. The value of the piggy bank increased deterministically at a rate of one euro per second, so that its value at time t was given by $V(t) = t$.

Payoffs were determined as follows. If one player pressed the button strictly before the other, the first mover received zero euros, while the second mover received the amount accumulated in the piggy bank at the time of the second pressed. If both players pressed the button simultaneously, the accumulated amount was divided equally between them. If neither player pressed the button within the 30-second limit, both players received zero euros.

Formally, the payoff function for player $i \in \{A, B\}$ (with $i \neq j$) is given by:

$$\pi_i(t_A, t_B) = \begin{cases} 0 & \text{if } t_i < t_j, \\ t_i & \text{if } t_i > t_j \text{ and } t_i \leq 30, \\ \frac{t_i}{2} & \text{if } t_i = t_j \leq 30, \\ 0 & \text{if } t_i = t_j = \infty. \end{cases}$$

CARDS and *PIGGY* were designed to elicit strategic reasoning under common knowledge of rationality, but they differ in whether dominance reasoning yields an immediate prediction. In *CARDS*, Player A's payoff satisfies $u_A(i, j) = i$ for all $j \in \{1, \dots, 5\}$, so choosing $i = 5$ is a strictly dominant action for Player A. By common knowledge of rationality, Player B can therefore infer that Player A will select $i = 5$ and best responds by choosing the highest value different from 5, namely $j = 4$, yielding the unique pure-strategy Nash equilibrium $(i, j) = (5, 4)$.

By contrast, in *PIGGY*, there is no strictly dominant action in the same sense, because payoffs hinge on being the *second* mover. Nevertheless, the game is finite-horizon with $t_i \in \{0, 1, \dots, 30\}$, which makes it amenable to backward reasoning. At the terminal second $t = 30$, pressing yields a weakly higher payoff than any earlier press given that the opponent can still press at 30: if both press at 30, the reward is split and each obtains $V(30)/2 = 15$, whereas pressing earlier while the opponent waits to 30 makes the deviator the first mover and yields zero. Iterating this argument backward implies that, under common knowledge of rationality, each player expects the other to wait until the last second, and therefore also waits until the last second. Accordingly, the game admits a (subgame-perfect) *NE* in pure strategies in which both players choose $t_A = t_B = 30$.

PIGGY is related to several strands of the literature. First, it is closely related to the original formulation of the Chicken Game (Rapoport & Chammah, 1966), capturing the strategic tension between acting early or waiting for the opponent. Second, it is connected to the War of Attrition framework (Fudenberg & Tirole, 1991; Smith, 1974), which emphasizes strategic delay, implicit costs of waiting, and the advantage of outlasting the opponent. Third, and more closely, our design relates to experimental studies of continuous-time waiting games with growing prizes and second-mover advantages (Friedman & Oprea, 2012; Oprea et al., 2011), which examine how subjects form higher-order beliefs and respond to strategic timing in-

centives.

Table 5 shows the composition of the sample for *PIGGY*, broken down by school academic year and gender.

Table 5: Number of participants in *PIGGY* by gender and school grade

Gender / School year	7th	8th	9th	10th	Total
Age assumed	11-12	12-13	14-15	16-17	
Male	77	93	69	82	321
Female	77	83	84	74	318
Other	1	4	3	6	14
% Female	49.68	46.11	53.85	45.68	48.70
Total	155	180	156	162	653

Note: These four academic years correspond to compulsory secondary education in Spain (*Educación Secundaria Obligatoria, ESO*), which students over 18 years of age cannot enroll in.

As shown by Brosig-Koch et al. (2015), children as young as six years old already possess the ability to reason backward. Accordingly, we expect children and adolescents in our sample (aged 10–17 years) have sufficiently developed cognitive skills. Nevertheless, despite Brocas and Carrillo (2020a) have shown that children from 5 years of age are able to understand and play equilibrium strategies in *BCG*, it is important to emphasize that strategic interaction in games is highly demanding, even for adults. For instance, Grosskopf and Nagel (2008) found that only about 10% of adults and 37% of participants at economics conferences were able to play the dominant strategy in *BCG*. Similar results were reported by Bosch-Rosa and Meissner (2020).

2.3 Hypotheses

2.3.1 Cognitive abilities

As previously documented in the literature, individuals with higher scores on cognitive ability tests are more likely to adopt dominant strategies. In particular, studies on *BCG* have shown that participants with higher cognitive skills tend to choose numbers closer to *NE* (Ballester et al., 2024; Gill et al., 2025), and even in one-shot settings (Burnham et al., 2009). This pattern is believed to arise from differences in participants’ ability to predict others’ decision-making processes (Brañas-Garza et al., 2012). Carpenter et al. (2013) also found a positive link between cognitive skills and playing optimal strategies in a game requiring induction, while Hermes and Schunk (2022) extracted similar conclusion to children who chose weakly dominated strategies.

For this reason:

Hypothesis 1: *CogAb has a positive effect with strategic thinking performance in CARDS and PIGGY.*

Cognitive abilities are especially relevant in the present study, as they reflect how participants solve strategic games and anticipate the behavior of other players. By including questions in our cognitive ability tests that capture reflectiveness and impulsivity, we assess participants’ capacity to anticipate others’ actions in order to select the best response. Accordingly, we expect that higher levels of reasoning will be associated with more strategic behavior, as individuals will anticipate that their counterparts are also more strategic. Consequently, we expect a positive relationship between the cognitive skills analyzed and the outcomes considered in our empirical analysis, particularly with respect to the rationality accuracy variable (*Rationality*).

2.3.2 Age

There is limited knowledge about how humans learn to make strategic decisions efficiently and how this ability develops during adolescence (Cooper & Weber, 2020). The question of how children and adolescents are able to play efficient strategies and how this ability evolves with age remains largely unresolved (Glätzle-Rützler et al., 2025).

Cognitive abilities develop with age (Brocas & Carrillo, 2018, 2020b, 2021b; Fe et al., 2022; Sher et al., 2014). However, the effects of age on the adoption of dominant strategies in economic games remain inconclusive. While Brocas and Carrillo (2021a) found that older adolescents coordinated more efficiently in a repeated coordination game, Glätzle-Rützler et al. (2025) reported no robust age effects in a one-shot experimental coordination game. Similarly, Brañas-Garza et al. (2026a) found no significant age effects in a one-shot experimental coordination game based on a Stag-Hunt played by children and adolescents.

Nevertheless, previous research has established that *ToM* develops around the age of six and enables individuals to take others’ perspectives, thereby facilitating more efficient and optimal decision-making (Grueneisen et al., 2015). Moreover, reasoning in economic games appears to stabilize during adolescence, as shown by Brosig-Koch et al. (2015) in an experiment involving a backward-induction task with a unique equilibrium, and by Brocas and Carrillo (2021b) in a dominance-solvable game. In addition, Czermak et al. (2016) found that strategic thinking stabilizes during adolescence in a series of normal-form games. Also, Hermes and Schunk (2022) found no age evidence in successful strategic interaction in children.

Therefore:

Hypothesis 2: *Age has no effect on the frequency of playing NE in CARDS and PIGGY.*

Given that our *CARDS* and *PIGGY* games are one-shot games, we expect our results to be more consistent with the findings of Glätzle-Rützler et al. (2025) and Brañas-Garza et al. (2026a) than with Brocas and Carrillo (2021a). We further expect strategic decision-making in our experiments to be stable across age groups, consistent with the findings reported in the aforementioned studies (Brocas & Carrillo, 2021b; Brosig-Koch et al., 2015; Czermak et al., 2016; Hermes & Schunk, 2022).

3 Methodology and results

3.1 Methodology

For the empirical analysis, we construct a set of outcome variables that capture participants' optimizing strategic behavior across *CARDS* and *PIGGY*. In each design, we define three complementary measures. Rather than focusing exclusively on whether individuals play the *NE*, we also consider the extent to which their behavior approximates it. This approach is motivated by evidence that subjects rarely play the *NE* exactly (Baddeley, 1966) and follows O'Neill (1991) in addressing both whether players reach equilibrium and how close they get to it.

To this end, we include two additional measures of deviation: *Trembling* and *Rationality*. *Trembling* captures small, potentially accidental deviations from the best response, consistent with the notion of *trembling hand* (Selten, 1983). Such deviations may arise from mistakes or from the cognitive demands of backward induction (*cognitive burden explanation*) (Gerber & Wichardt, 2010; Palacios-Huerta & Volij, 2009), as well as from limited expectations about others' rationality (*common knowledge breakdown*) (Binmore, 1987). In turn, *Rationality* provides a continuous measure of how closely participants' decisions align with the *NE*.

1. *Nash*

- In *CARDS*: binary variable equal to 1 if participants selected card number 5, and 0 otherwise. Since Player A's payoff satisfies $\pi_A(i, j) = i$ for all $j \in \{1, \dots, 5\}$, choosing $i = 5$ is the strictly dominant action and corresponds to the *NE*.
- In *PIGGY*: binary variable equal to 1 if the piggy bank was pressed at second 30, and 0 otherwise (excluding choices above 30). Under common knowledge of rationality, both players optimally wait until the last second, yielding a pure-strategy *NE* at $t_A = t_B = 30$.

2. *Trembling*

- In *CARDS*: binary variable equal to 1 if participants selected cards 5 or 4, and 0 otherwise, allowing for small deviations from the optimal choice.
- In *PIGGY*: binary variable equal to 1 if participants pressed the piggy bank between seconds 28 and 30 (inclusive). This window increases the likelihood of capturing near-equilibrium behavior⁷.

3. *Rationality*

- In *CARDS*: continuous measure based on the selected card number, where higher values indicate choices closer to the *NE*. The variable is standardized between 0 and 1.
- In *PIGGY*: continuous measure based on the stopping time, with later decisions reflecting higher rationality. This variable is also standardized between 0 and 1.

⁷Results are qualitatively similar when alternative time windows are used.

To examine the determinants of such behavior, we include a series of independent and control variables capturing individual cognitive performance, elicitation of preferences and demographic characteristics.

Our main measure of cognitive ability (*CogAb*) is an age-adjusted index expressed in standard deviation units, constructed separately for each experimental subsample (*CogAb_std - CARDS* and *CogAb_std - PIGGY*) to account for residual age effects. The composite index combines performance across three domains: cognitive reflection (measured through a short version of the Cognitive Reflection Test, *CRT* (Frederick, 2005; Thomson & Oppenheimer, 2016)), financial literacy (*FinAb*; Alfonso et al. (2025)), and probabilistic reasoning (*Del*; Delavande and Rohwedder (2008)), with participants receiving one point per correct response. In addition, we control for time preferences using *PatienceS*, defined as the standardized number of patient choices in *The Truck Task*, a visual adaptation of the Multiple Price List (Alfonso et al., 2023a; Coller & Williams, 1999), where higher values indicate greater willingness to delay gratification. Finally, risk preferences are captured by *RiskyS*, the standardized number of risky choices in *The Gumball Machine Task*, a graphical version of the Holt and Laury (2002) test (Vasco & De Francisco, 2023), with values ranging from 0 to 1.

3.2 Results

The statistical analysis presented in this section relies on linear regression models with robust standard errors clustered at the class level and school fixed effects. Results are reported in terms of p-values in the main text, while complete tables with coefficients and standard errors are provided in the Appendix. The dependent variables mentioned before correspond to equilibrium-consistent behavior for both experimental designs.

All linear regressions include as main explanatory variables our measures of cognitive ability (*CogAb*), time preferences (*PatienceS*), and risk preferences (*#RiskyS*). In addition, we control for participants' age (measured in years) and gender through a dummy variable (*Female*). We also include a dummy variable (*Repeater*) indicating whether the student has repeated a grade, and *Class size* (number of students) to account for classroom characteristics. Finally, for *PIGGY* only, we include a variable capturing potential order effects (*Second Position*), equal to 1 if the outcome was elicited in second place, as this design involves two sequential tasks.

3.2.1 Result I - Strategic behavior across designs

Figure 1 presents the distribution of optimization behavior across both experimental designs. Optimization in each design corresponds to the main outcome variables used in the analysis, described in the previous section (*Nash*, *Trembling* and *Rationality*).

Clear differences in strategic performance emerge across tasks. In *CARDS*, 26.3% of participants play *NE*. In contrast, only 4.3% of participants in *PIGGY*

play the best response, reflecting the complexity of the design structure and the deeper level of reasoning it requires, related to a game where time is employed as a determinant factor of rationality.

The introduction of *Trembling* as outcome shows that the number of participants who play this strategy increases from *Nash*, reaching 53.5% and 6.6% of participants in *CARDS* and *PIGGY*, respectively. Therefore, introducing *Trembling* as a variable in the analysis captures a larger proportion of subjects who deviate from the *NE*. We know that deviating from the *NE* is risky because it would result in a lower payoff. Following Nowik (2016), a player may nevertheless attempt to take advantage if they suspect that the other player is not playing the *NE*, or the deviation may occur by accident, as Selten (1983) describes through the *trembling hand* concept. Moreover, such behavior could also be driven by a lack of motivation or difficulty in the strategic reasoning required to fully understand the game, commonly referred to as the cognitive burden explanation (Gerber & Wichardt, 2010; Palacios-Huerta & Volij, 2009). Finally, it may stem from players not expecting their counterparts to be fully rational, a phenomenon known as the common knowledge breakdown explanation (Binmore, 1987).

Finally, for *Rationality* we see a complete different image across experiments. For *CARDS*, we observe a positive slope, where participants understood they had to choose higher numbers than lower. Nevertheless, in *PIGGY* we observe three key points: at the beginning, at the middle, and at the end of the time countdown. Indeed, we find the highest percentage of participants that pressed the piggybank between seconds 15 and 18, saying that they expected the rival pressed before that time⁸.

These results highlight meaningful variation in optimization behavior across settings with different cognitive and strategic demands. While *CARDS* appears to facilitate strategic performance through a salient focal point, *PIGGY* requires higher-order reasoning and dynamic decision-making, leading to lower observed optimization rates.

This bring us to our first result:

Result 1: *Subjects play more frequently NE in CARDS (static) than in PIGGY (dynamic).*

To provide a more robust conclusion, the next subsection presents a linear regression analysis of optimization behavior by cognition. This allows us to assess the statistical significance through cognitive abilities in explaining strategic decision-making.

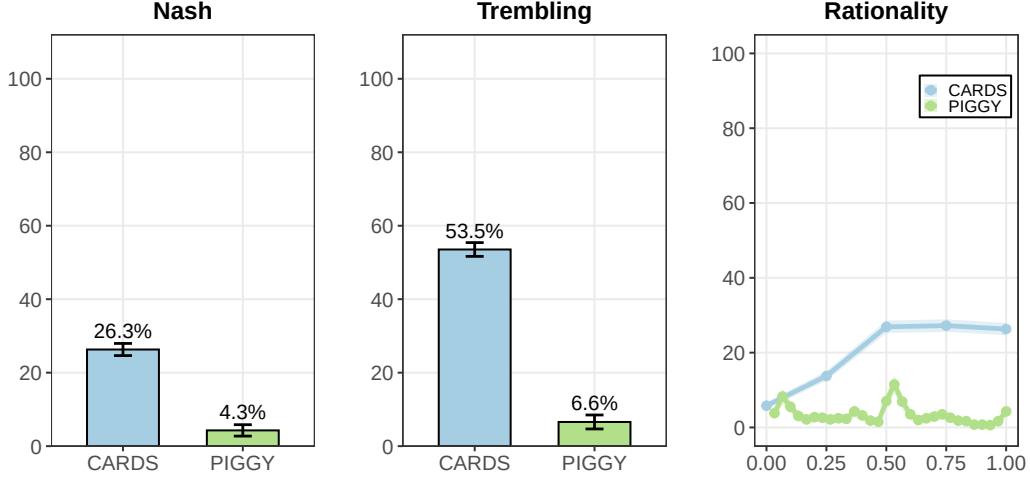
3.2.2 Result II - Determinants of the optimization

To explore developmental patterns in strategic reasoning, we examine the extent to which participants optimized their decisions⁹ as a function of their cognitive abilities.

⁸See in Table 6 in Appendix the distribution of responses.

⁹See in Table 6 the distribution of responses in both games.

Figure 1: Distribution of optimization by experimental design



Note: Bars and functions represent the percentage of subjects optimizing their decisions in *CARDS* and *PIGGY*.

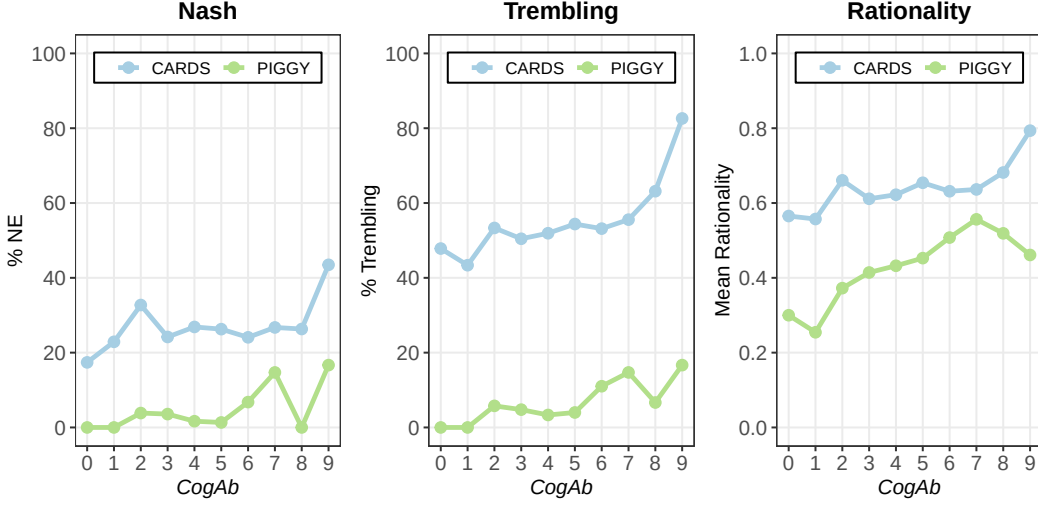
Figure 2 displays, for each level of *CogAb*, the average share of participants playing *Nash*, *Trembling* and the mean *Rationality*, separately for *CARDS* (blue) and *PIGGY* (green)¹⁰.

Several patterns emerge from the figure. First, *Nash* remains low across all cognitive ability levels in both games, with *CARDS* consistently above *PIGGY*. Neither line displays a clear monotonic trend, though there is a mild upward tendency at the highest *CogAb* scores. Second, *Trembling* rates tell a strikingly different story across games. In *CARDS*, a large share of participants play *Trembling* throughout the cognitive ability distribution, hovering around 50% for most levels and rising sharply to approximately 80% at the highest scores. In contrast, *Trembling* in *PIGGY* remains near zero across virtually all *CogAb* levels, pointing to fundamentally different behavioral patterns between the static and dynamic tasks. Third, the right panel confirms that rationality scores are consistently higher in *CARDS* than in *PIGGY* at every level of cognitive ability, and both series display a positive gradient with *CogAb*. Specifically, mean rationality in *CARDS* rises from approximately 0.55 at the lowest scores to around 0.80 at *CogAb* = 9, while in *PIGGY* the corresponding range is roughly 0.30 to 0.45.

Taken together, these results suggest that cognitive ability is positively associated with strategic sophistication in both games, but that the mapping between cognitive resources and equilibrium play is considerably stronger (and more aligned with the *Trembling*) in the static card game than in the dynamic timing task.

¹⁰See in Figure 4 the distribution of responses of *CARDS* and *PIGGY* by *CRT*, *FinAb* and *Del*.

Figure 2: Strategic performance by *CogAb*



Note: Mean strategic performance in *CARDS* (blue) and *PIGGY* (green) as a function of *CogAb* score (0–9). The left panel shows the percentage of participants playing a Nash Equilibrium strategy; the center panel shows the percentage playing a Trembling Hand Equilibrium strategy; the right panel displays the mean rationality score, normalized to $[0, 1]$ so that both games are expressed on a common scale.

Before turning to the regression analysis, we examine pairwise correlations between cognitive ability and the different measures of strategic behavior. In *CARDS*, cognitive ability is weakly correlated with *Nash* ($r = 0.0089$, $p > 0.1$), but shows a small and significant association with *Trembling* ($r = 0.0623$, $p < 0.05$) and *Rationality* ($r = 0.0554$, $p < 0.05$). In contrast, in *PIGGY* the correlations are stronger across all outcomes: cognitive ability is positively associated with *Nash* ($r = 0.1069$, $p < 0.01$), *Trembling* ($r = 0.1124$, $p < 0.01$), and especially *Rationality* ($r = 0.2167$, $p < 0.01$). Additionally, the outcome variables are strongly correlated among themselves in both designs, particularly between *Trembling* and *Rationality*, suggesting that they capture closely related dimensions of strategic behavior. These descriptive patterns provide initial evidence of a stronger link between cognitive ability and strategic performance in *PIGGY*.

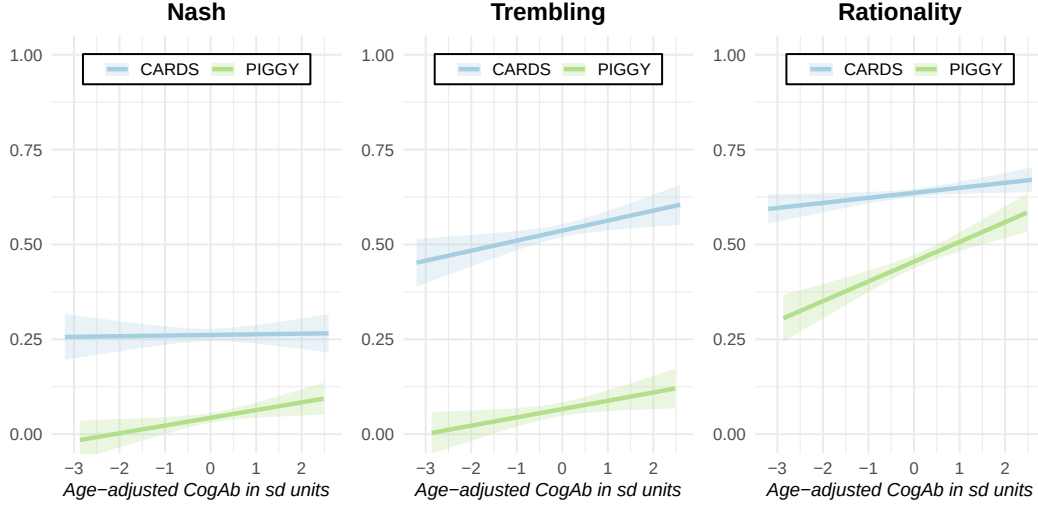
Figure 3 displays columns 2, 4, 6, 8, 10 and 12 of Table 8 (see in the Appendix), corresponding to the fully-specified models that include all control variables. It illustrates the relationship between cognitive abilities and the probability of optimizing in each experimental design, using linear regression models¹¹.

The figure reveals three distinct patterns. First, *Nash* shows no meaningful relationship with cognitive ability in either game: both lines are essentially flat across the full distribution of age-adjusted *CogAb*. Second, *Trembling* presents a sharp contrast between designs. In *CARDS*, predicted it is substantially higher, and exhibits a mild positive gradient, whereas in *PIGGY* it remains close to zero throughout, reinforcing the idea that the static and dynamic tasks elicit fundamentally different behavioral responses. Third, and most strikingly, *Rationality* displays a clear positive slope for both games: higher cognitive ability is associated with greater

¹¹See in tables 9 and 10 the determinants of *Nash*, *Trembling* and *Rationality* for *CRT*, *FinAb* and *Del* in *CARDS* and *PIGGY*, respectively.

rationality regardless of the experimental design.

Figure 3: Cognitive ability and strategic performance: linear regression estimates



Note: Estimation of the impact of age-adjusted cognitive ability index in standard deviation units for each experimental design subsample in optimizing through linear regressions. For all models we regress the effect of different control variables respect to outcomes (*Nash*, *Trembling* and *Rationality*). Control variables include age of participants and gender (*Female*), the standardized measures of risk attitudes (*RiskyS*) and patience (*PatienceS*), if they had to enroll at least twice in the same school year because they failed (*Repeater*) and the class size. Also, to control potential order effects in *PIGGY*, we include the variable *SecondPosition*, which indicates whether *PIGGY* design question was presented in second position of the design. This variable is included only in the regressions related to *PIGGY*. All specifications also include school fixed effects, and all regressions are estimated using heteroskedasticity-robust standard errors clustered at the class level.

Table 8 reports the estimated coefficients for all linear regression models used in the analysis. For each optimization variable, two specifications are presented, with the only difference to incorporate in the second one the age adjusted cognitive ability score¹². All specifications include the linear effect of control variables respectively outcome variables in each experimental design (*Nash*, *Trembling* and *Rationality*). While models (1) - (6) explore the relationship between control variables and the outcomes in *CARDS*, models (7) - (12) show the specification across controls and outcomes in *PIGGY*.

1. *CARDS*:

For *Nash* in *CARDS*, we do not observe a statistically significant relationship between cognitive ability and the likelihood of optimizing (see Table 8 in the Appendix). The *p*-value associated with *CogAb_std* is 0.861 in model (2).

¹²To isolate individual differences in cognitive ability net of age, we residualized the composite cognitive score (in *CogAb*) with respect to participants' age within the analytic subsample. First, we estimated a linear regression of the *CogAb* on age. After estimating the model, we computed the regression residuals for each individual. These residuals, represent the part of *CogAb* that is not explained by age. Finally, to facilitate interpretation and comparability across analyses, we standardized the residualized cognitive measure to have mean 0 and standard deviation 1 within the subsample. The resulting variable can be interpreted as an age-adjusted cognitive ability index in standard deviation units: a value of $+1$ indicates the participant's cognitive score is one standard deviation above the average after removing the linear association with age.

Nevertheless, when we introduce the rest of outcomes, the cognitive ability specifications become significant in both models (4) and (6), reporting significant and positive associations with $p = 0.008$ and $p = 0.028$ for *Trembling* and *Rationality*, respectively.

For the rest of the control variables analyzed in models (1) - (6), we do observe a marginally positive relationship between *PatienceS*, but it becomes less significant as we introduce and change the outcome of optimization, reporting a $p = 0.056$ in the sixth specification. Lastly, we do not observe gender or age effect in any of the first six linear regressions models.

2. *PIGGY*:

However, in *PIGGY*, we do observe a statistically significant relationship between cognitive ability and *Nash*, with a p -value of 0.023. As with *CARDS*, when we analyze the rest of the outcomes in models (10) and (12), we report a significant and positive relationship between *CogAb* and the capacity to optimize, with $p = 0.029$ and $p = 0.000$ for *Trembling* and *Rationality*, respectively.

In the rest of the control variables we observe a negative significant relationship between *Trembling* ($p = 0.068$) in model (9) and *Rationality* and being a grade repeater ($p = 0.002$ and $p = 0.047$ in models (11) and (12), respectively). Furthermore, we do not observe any significant age, gender and order effects for *PIGGY* in any of the outcomes considered, with the exception with a marginal significance between *Trembling* and *RiskyS* in model (9) ($p = 0.096$)¹³.

Therefore, this bring us to the two following results:

Result 2a: *Cognitive abilities significantly explain the frequency of playing NE (or close to NE) in both CARDS and PIGGY, and are more sensitive predictors of strategic behavior in PIGGY than in CARDS, particularly with respect to the accuracy component of Rationality.*

Result 2b: *There is no age effect on the ability to think strategically among adolescents.*

As reported in the previous literature, age does not significantly affect the probability of playing the equilibrium (Brañas-Garza et al., 2026a; Brocas & Carrillo, 2021b; Brosig-Koch et al., 2015; Czermak et al., 2016; Glätzle-Rützler et al., 2025; Hermes & Schunk, 2022). As indicated in other studies, strategic sophistication is closely tied to the level of cognitive ability (Ballester et al., 2024; Brañas-Garza et al., 2012, 2026a; Brocas & Carrillo, 2018; Carpenter et al., 2013; Fe et al., 2022; Gill et al., 2025; Hermes & Schunk, 2022). This result holds across all model specifications for *Trembling* and *Rationality*, becoming even stronger in this outcome in *PIGGY*.

¹³See in Appendix as robustness check, the examination of seconds to pulse in *PIGGY* differ by device. Table 11 shows that no statistically significant differences across devices. These results hold both when including school fixed effects, with class fixed effects and with no fixed effects (with the exception of a marginal significance in model (4) ($p = 0.084$), suggesting that the device used to complete the experiment does not meaningfully affect response times.

4 Conclusion

In this paper, we analyze the determinants of cognitive abilities in the strategic behavior of adolescents using two experimental games: *CARDS* and *PIGGY*.

Both consisted of one-shot, two-player versions with complete and symmetric information. *CARDS* is an adaptation of the *BCG* (Nagel, 1995) implemented through a card game, in which participants had to choose one card from a set of five. *PIGGY* is a reaction-time competition task inspired by the War of Attrition framework (Smith, 1974) and the original formulation of the Chicken Game (Rapoport & Chammah, 1966), incorporating a dynamic accumulation process with a finite horizon and strategic incentives driven by waiting and second-mover advantages, in which participants had to press a virtual piggy bank before the 30-second countdown elapsed.

Participants completed only one of the two experimental designs, and incentives were hypothetical. We are aware that the literature is not unanimous on this point: while several studies find no significant differences between paid and unpaid conditions (Brañas-Garza et al., 2021, 2023, 2026b), others document meaningful incentive effects, particularly in risk-taking and effort tasks (Camerer & Hogarth, 1999; Holt & Laury, 2002; Xu et al., 2018). Adolescents in particular may be more responsive to real monetary stakes in such domains (Kray et al., 2018). Importantly, strategic coordination games like those used here are among the domains where such differences are least likely to emerge, and our design follows a validated approach for adolescent samples.

To measure cognitive abilities, participants completed three assessments that form the synthetic index *CogAb*: a Cognitive Reflection Test (Frederick, 2005; Thomson & Oppenheimer, 2016), a financial literacy test (Alfonso et al., 2025), and a probabilistic reasoning task (Delavande & Rohwedder, 2008).

To capture strategic optimization, we define three outcomes for each game: *Nash*, *Trembling*, and *Rationality*. Rather than focusing solely on exact equilibrium play (Baddeley, 1966; O'Neill, 1991), *Trembling* captures near-best-response behavior arising from cognitive demands or limited expectations about others' rationality (Binmore, 1987; Gerber & Wichardt, 2010), while *Rationality* provides a continuous measure of proximity to the Nash Equilibrium (Selten, 1983).

Our study yields four main findings. First, subjects play *Nash* more frequently in *CARDS* than in *PIGGY*. Second, strategic thinking depends on the cognitive abilities of children and adolescents across both experimental designs. Third, cognitive abilities are more strongly related to strategic thinking in the dynamic game (*PIGGY*) than in the static one (*CARDS*), and these correlations are more closely tied to the rationality accuracy variable, which indicates how closely play *Nash*. Finally, we do not find evidence that age explains an evolutionary pattern in the development and performance of strategic decision-making in either game.

This study contributes to the growing literature on experimental research with

children and adolescents by advancing both methodological and theoretical understanding of strategic behaviour during this critical developmental stage (Brocas & Carrillo, 2020a; Sutter et al., 2019). It provides evidence that strategic sophistication in adolescence is closely tied to cognitive reflection rather than to age or physical maturation, offering new insight into how reasoning and anticipation develop in parallel with cognitive abilities (Ballester et al., 2024; Brañas-Garza et al., 2012, 2026a; Brocas & Carrillo, 2018; Carpenter et al., 2013; Fe et al., 2022; Gill et al., 2025; Hermes & Schunk, 2022). It underscores that well-calibrated experimental tools are essential not only for observing behaviour but for understanding the cognitive processes that underpin it. Additionally, we contribute to the literature by introducing two novel experimental games that, to our knowledge, have not been previously analysed or implemented. Our results show that one of them (*PIGGY*) features a design that is particularly informative in its relationship with cognition among children and adolescents, while *CARDS* also reveals a strong association between strategic performance and cognitive abilities. Lastly, the absence of a strong age effect on playing the best response is consistent with findings from previous studies (Brañas-Garza et al., 2026a; Brocas & Carrillo, 2021b; Brosig-Koch et al., 2015; Czermak et al., 2016; Glätzle-Rützler et al., 2025; Hermes & Schunk, 2022).

Some limitations of this study should be acknowledged. First, as noted above, incentives were hypothetical throughout the experiment. Although we rely on a well-established literature suggesting that monetary incentives do not systematically alter behavior in strategic tasks (Brañas-Garza et al., 2021, 2023, 2026b), this remains a potential concern. Second, the experimental design was between-subjects, meaning that different participants completed each game. While this avoids potential learning and carryover effects, it prevents direct within-individual comparisons across *CARDS* and *PIGGY*, making it difficult to disentangle whether observed differences reflect properties of the games themselves or pre-existing differences between participant groups. Third, the two games employed were selected to represent static and dynamic strategic environments, respectively, but they differ along multiple dimensions beyond their temporal structure. As a result, any differences in the predictive power of cognitive abilities across games cannot be straightforwardly attributed to the static versus dynamic distinction, and caution is warranted when generalizing these findings to other strategic settings.

Nevertheless, other strands of the literature suggest that the ability to reason abstractly—such as hypothetical and counterfactual reasoning—fully develops during middle school (Piaget, 2008; Rafetseder et al., 2013). As individuals in our sample grow older, they may therefore find it easier to construct counterfactuals about their counterparts’ behavior. However, this ability appears to manifest primarily in intelligence and logic tests and does not necessarily transfer to strategic interaction games. Looking ahead, future research could extend these designs to different social contexts or to younger age groups, thereby testing the robustness of the observed relationships across developmental and cultural settings. Moreover, integrating short comprehension surveys within the experimental protocols could help assess participants’ understanding of the tasks in real time, providing an additional methodological layer for validating behavioural data and refining experimental de-

signs.

References

- Alfonso, A., Brañas-Garza, P., Jorrat, D., Lomas, P., Prissé, B., Vasco, M., & Vázquez-De Francisco, M. J. (2023a). The adventure of running experiments with teenagers. *Journal of Behavioral and Experimental Economics*, 106, 102048.
- Alfonso, A., Brañas-Garza, P., Jorrat, D., Prissé, B., & Vázquez-De Francisco, M. J. (2025). The baking of preferences throughout the high school. *Journal of Behavioral and Experimental Economics*, 102396.
- Baddeley, A. D. (1966). The capacity for generating information by randomization. *The Quarterly Journal of Experimental Psychology*, 18(2), 119–129.
- Ballester, C., Rodriguez-Moral, A., & Vorsatz, M. (2024). Cognitive reflection in experimental anchored guessing games. *Games and Economic Behavior*, 148, 179–195.
- Baron-Cohen, S., O’riordan, M., Stone, V., Jones, R., & Plaisted, K. (1999). Recognition of faux pas by normally developing children and children with asperger syndrome or high-functioning autism. *Journal of Autism and Developmental Disorders*, 29(5), 407–418.
- Binmore, K. (1987). Modeling rational players: Part i. *Economics & Philosophy*, 3(2), 179–214.
- Birch, S., Li, V., Haddock, T., Ghrear, S., Brosseau-Liard, P., Baimel, A., & Whyte, M. (2017). Perspectives on perspective taking: How children think about the minds of others. *Advances in Child Development and Behavior*, 52, 185–226.
- Bornstein, G., Kugler, T., & Ziegelmeyer, A. (2004). Individual and group decisions in the centipede game: Are groups more “rational” players? *Journal of Experimental Social Psychology*, 40(5), 599–605.
- Bosch-Rosa, C., & Meissner, T. (2020). The one player guessing game: A diagnosis on the relationship between equilibrium play, beliefs, and best responses. *Experimental Economics*, 23(4), 1129–1147.
- Brañas-Garza, P., Estepa, L., Jorrat, D., Orozco-Olvera, V., & Rascon Ramirez, E. G. (2021). To pay or not to pay: Measuring risk preferences in lab and field. *Judgment and Decision Making*, 16(5), 1290–1313.
- Brañas-Garza, P., García-Muñoz, T., & González, R. H. (2012). Cognitive effort in the beauty contest game. *Journal of Economic Behavior & Organization*, 83(2), 254–260.
- Brañas-Garza, P., Jorrat, D., Espín, A. M., & Sánchez, A. (2023). Paid and hypothetical time preferences are the same: Lab, field and online evidence. *Experimental Economics*, 26(2), 412–434.

- Brañas-Garza, P., Jorrat, D., & Lomas, P. (2026a). *Asymmetric development of strategic coordination across the life cycle* [Mimeo].
- Brañas-Garza, P., Jorrat, D., & Lomas, P. (2026b). *Coordination with and without incentives* [Manuscript under review].
- Brandts, J., & Cooper, D. J. (2006). A change would do you good.... an experimental study on how to overcome coordination failure in organizations. *American Economic Review*, 96(3), 669–693.
- Brocas, I., & Carrillo, J. D. (2018). The determinants of strategic thinking in preschool children. *PLoS ONE*, 13(5), e0195456.
- Brocas, I., & Carrillo, J. D. (2020a). The evolution of choice and learning in the two-person beauty contest game from kindergarten to adulthood. *Games and Economic Behavior*, 120, 132–143.
- Brocas, I., & Carrillo, J. D. (2020b). Introduction to special issue “understanding cognition and decision making by children.” studying decision-making in children: Challenges and opportunities. *Journal of Economic Behavior & Organization*, 179, 777–783.
- Brocas, I., & Carrillo, J. D. (2020c). Iterative dominance in young children: Experimental evidence in simple two-person games. *Journal of Economic Behavior & Organization*, 179, 623–637.
- Brocas, I., & Carrillo, J. D. (2021a). Dynamic coordination in efficient and fair strategies: A developmental perspective. CEPR Discussion Paper No. DP16683.
- Brocas, I., & Carrillo, J. D. (2021b). Steps of reasoning in children and adolescents. *Journal of Political Economy*, 129(7), 2067–2111.
- Brosig-Koch, J., Heinrich, T., & Helbach, C. (2015). Exploring the capability to reason backwards: An experimental study with children, adolescents, and young adults. *European Economic Review*, 74, 286–302.
- Burnham, T. C., Cesarini, D., Johannesson, M., Lichtenstein, P., & Wallace, B. (2009). Higher cognitive ability is associated with lower entries in a p-beauty contest. *Journal of Economic Behavior & Organization*, 72(1), 171–175.
- Camerer, C. F. (2003). *Behavioral game theory: Experiments in strategic interaction*. Princeton University Press.
- Camerer, C. F., & Hogarth, R. M. (1999). The effects of financial incentives in experiments: A review and capital-labor-production framework. *Journal of Risk and Uncertainty*, 19(1), 7–42.
- Carlson, S. M., & Moses, L. J. (2001). Individual differences in inhibitory control and children’s theory of mind. *Child Development*, 72(4), 1032–1053.
- Carlson, S. M., Moses, L. J., & Claxton, L. J. (2004). Individual differences in executive functioning and theory of mind: An investigation of inhibitory control and planning ability. *Journal of Experimental Child Psychology*, 87(4), 299–319.

- Carpenter, J., Graham, M., & Wolf, J. (2013). Cognitive ability and strategic sophistication. *Games and Economic Behavior*, 80, 115–130.
- Cawley, J., Heckman, J., & Vytlačil, E. (2001). Three observations on wages and measured cognitive ability. *Labour Economics*, 8(4), 419–442.
- Coller, M., & Williams, M. B. (1999). Eliciting individual discount rates. *Experimental Economics*, 2(2), 107–127.
- Cooper, D. J., & Weber, R. A. (2020). Recent advances in experimental coordination games. In C. M. Capra, R. T. A. Croson, M. L. Rigdon, & T. S. Rosenblat (Eds.), *Handbook of experimental game theory* (pp. 149–183). Edward Elgar Publishing.
- Crawford, V. P., Costa-Gomes, M. A., & Iriberri, N. (2013). Structural models of nonequilibrium strategic thinking: Theory, evidence, and applications. *Journal of Economic Literature*, 51(1), 5–62.
- Crone, E. A., & Dahl, R. E. (2012). Understanding adolescence as a period of social-affective engagement and goal flexibility. *Nature Reviews Neuroscience*, 13(9), 636–650.
- Czermak, S., Feri, F., Glätzle-Rützler, D., & Sutter, M. (2016). How strategic are children and adolescents? experimental evidence from normal-form games. *Journal of Economic Behavior & Organization*, 128, 265–285.
- Davis, H. L., & Pratt, C. (1995). The development of children’s theory of mind: The working memory explanation. *Australian Journal of Psychology*, 47(1), 25–31.
- Delavande, A., & Rohwedder, S. (2008). Eliciting subjective probabilities in internet surveys. *Public Opinion Quarterly*, 72(5), 866–891.
- Drobner, C., Goerg, S. J., & Kopsacheilis, O. (2024). *Real incentives really matter* (SSRN Working Paper No. 5013465). <https://ssrn.com/abstract=5013465>
- Egan, D., Merkle, C., & Weber, M. (2014). Second-order beliefs and the individual investor. *Journal of Economic Behavior & Organization*, 107, 652–666.
- Fe, E., Gill, D., & Prowse, V. (2022). Cognitive skills, strategic sophistication, and life outcomes. *Journal of Political Economy*, 130(10), 2643–2704.
- Fey, M., McKelvey, R. D., & Palfrey, T. R. (1996). An experimental study of constant-sum centipede games. *International Journal of Game Theory*, 25(3), 269–287.
- Finucane, M. L., & Gullion, C. M. (2010). Developing a tool for measuring the decision-making competence of older adults. *Psychology and Aging*, 25(2), 271.
- Flynn, E. (2007). The role of inhibitory control in false belief understanding. *Infant and Child Development*, 16(1), 53–69.
- Frederick, S. (2005). Cognitive reflection and decision making. *Journal of Economic Perspectives*, 19(4), 25–42.
- Friedman, D., & Oprea, R. (2012). A continuous dilemma. *American Economic Review*, 102(1), 337–363.

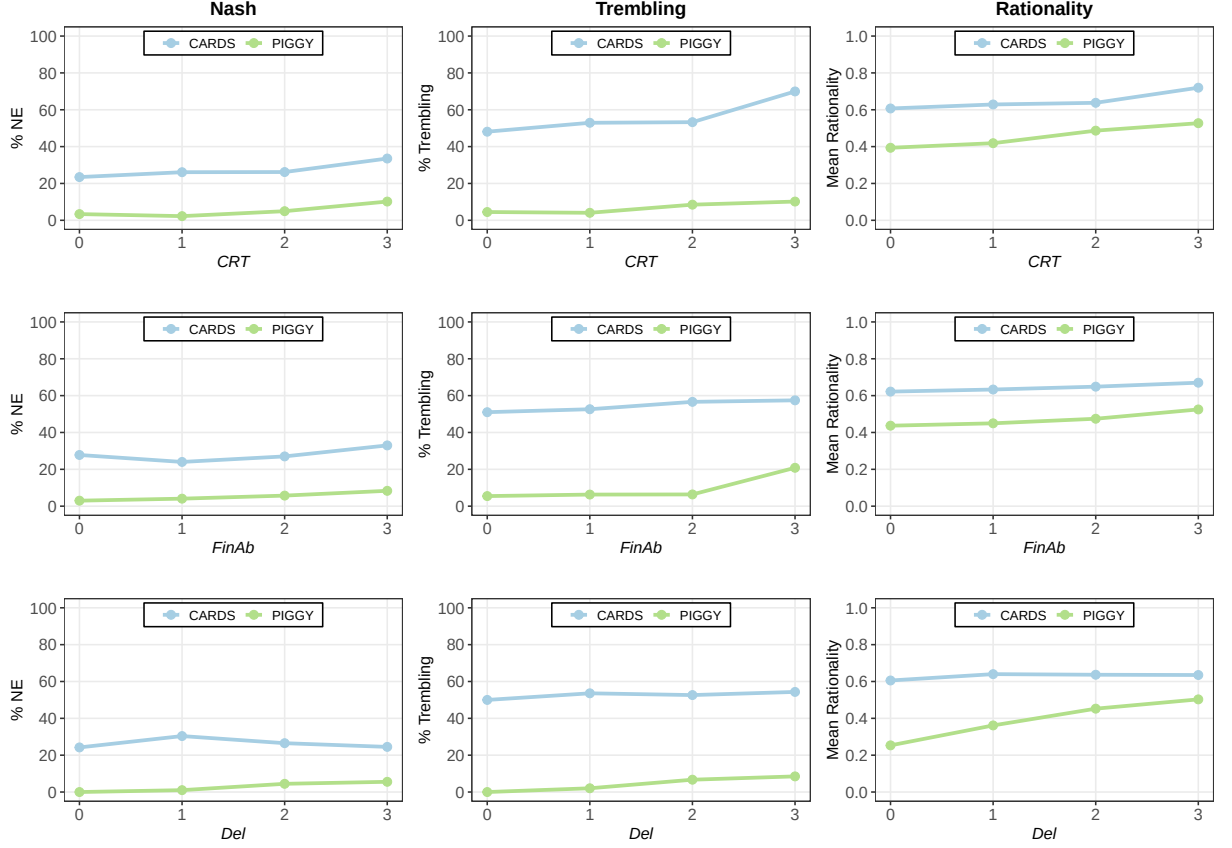
- Fudenberg, D., & Tirole, J. (1991). *Game theory*. MIT Press.
- Gerber, A., & Wichardt, P. C. (2010). Iterated reasoning and welfare-enhancing instruments in the centipede game. *Journal of Economic Behavior & Organization*, 74(1-2), 123–136.
- Gill, D., Knepper, Z., Prowse, V., & Zhou, J. (2025). How cognitive skills affect strategic behavior: Cognitive ability, fluid intelligence and judgment. *Games and Economic Behavior*, 149, 82–95.
- Glätzle-Rützler, D., Sutter, M., & Zoller, C. (2025). Coordination games played by children and teenagers: On the influence of age, group size, and incentives. *Experimental Economics*, 28(2), 217–236.
- Grosskopf, B., & Nagel, R. (2008). The two-person beauty contest. *Games and Economic Behavior*, 62(1), 93–99.
- Grueneisen, S., Wyman, E., & Tomasello, M. (2015). Children use salience to solve coordination problems. *Developmental Science*, 18(3), 495–501.
- Hala, S., Hug, S., & Henderson, A. (2003). Executive function and false-belief understanding in preschool children: Two tasks are harder than one. *Journal of Cognition and Development*, 4(3), 275–298.
- Harbaugh, W. T., & Krause, K. (2000). Children’s altruism in public good and dictator experiments. *Economic Inquiry*, 38(1), 95–109.
- Harbaugh, W. T., Krause, K., Liday, S. G., & Vesterlund, L. (2003). Trust in children. *Trust and reciprocity: interdisciplinary lessons from experimental research*, 302, 322.
- Heckman, J. J., Stixrud, J., & Urzua, S. (2006). The effects of cognitive and noncognitive abilities on labor market outcomes and social behavior. *Journal of Labor Economics*, 24(3), 411–482.
- Hermes, H., & Schunk, D. (2022). If you could read my mind—an experimental beauty-contest game with children. *Experimental Economics*, 25(1), 229–253.
- Holt, C. A., & Laury, S. K. (2002). Risk aversion and incentive effects. *American Economic Review*, 92(5), 1644–1655. <https://doi.org/10.1257/000282802762024700>
- Hughes, C. (1998). Executive function in preschoolers: Links with theory of mind and verbal ability. *British Journal of Developmental Psychology*, 16(2), 233–253.
- Kautz, T., Heckman, J. J., Diris, R., ter Weel, B., & Borghans, L. (2014). *Fostering and measuring skills: Improving cognitive and non-cognitive skills to promote lifetime success* (NBER Working Paper No. 20749). National Bureau of Economic Research. Cambridge, MA.
- Kray, J., Schmitt, H., Lorenz, C., & Ferdinand, N. K. (2018). The influence of different kinds of incentives on decision-making and cognitive control in adolescent development: A review of behavioral and neuroscientific studies. *Frontiers in Psychology*, 9, 768. <https://doi.org/10.3389/fpsyg.2018.00768>

- Landry, S. H., Miller-Loncar, C. L., Smith, K. E., & Swank, P. R. (2002). The role of early parenting in children's development of executive processes. *Developmental Neuropsychology*, 21(1), 15–41.
- Levitt, S. D., List, J. A., & Sadoff, S. E. (2011). Checkmate: Exploring backward induction among chess players. *American Economic Review*, 101(2), 975–990.
- Masclet, D., & Rebière, T. (2021). Do real stakes always matter? experimental evidence on social and anti-social preferences. *Experimental Evidence on Social and Anti-Social Preferences*.
- McKelvey, R. D., & Palfrey, T. R. (1992). An experimental study of the centipede game. *Econometrica*, 60(4), 803–836.
- Moses, L. J., & Carlson, S. M. (2004). Self-regulation and children's theories of mind. In C. Lightfoot, C. Lalonde, & M. Chandler (Eds.), *Changing conceptions of psychological life* (pp. 139–158). Lawrence Erlbaum Associates.
- Müller, U., Liebermann-Finestone, D. P., Carpendale, J. I., Hammond, S. I., & Bibok, M. B. (2012). Knowing minds, controlling actions: The developmental relations between theory of mind and executive function from 2 to 4 years of age. *Journal of Experimental Child Psychology*, 111(2), 331–348.
- Murnighan, J. K., & Saxon, M. S. (1998). Ultimatum bargaining by children and adults. *Journal of Economic Psychology*, 19(4), 415–445.
- Nagel, R. (1995). Unraveling in guessing games: An experimental study. *American Economic Review*, 85(5), 1313–1326.
- Nowik, I. (2016). The game on the risk in deviating from nash equilibrium. *2016 Second International Symposium on Stochastic Models in Reliability Engineering, Life Science and Operations Management (SMRLO)*, 263–273.
- Oechssler, J., Roider, A., & Schmitz, P. W. (2009). Cognitive abilities and behavioral biases. *Journal of Economic Behavior & Organization*, 72(1), 147–152.
- O'Neill, B. (1991). Comments on brown and rosenthal's reexamination. *Econometrica*, 59(2), 503–507.
- Oprea, R., Henwood, K., & Friedman, D. (2011). Separating the hawks from the doves: Evidence from continuous time laboratory games. *Journal of Economic Theory*, 146(6), 2206–2225.
- Palacios-Huerta, I., & Volij, O. (2009). Field centipedes. *American Economic Review*, 99(4), 1619–1635.
- Piaget, J. (2008). Intellectual evolution from adolescence to adulthood. *Human Development*, 51(1), 40–47.
- Ponti, G. (2000). Cycles of learning in the centipede game. *Games and Economic Behavior*, 30(1), 115–141.
- Pulford, B. D., Colman, A. M., Lawrence, C. L., & Krockow, E. M. (2017). Reasons for cooperating in repeated interactions: Social value orientations, fuzzy traces, reciprocity, and activity bias. *Decision*, 4(2), 102–122.

- Rafetseder, E., Schwitalla, M., & Perner, J. (2013). Counterfactual reasoning: From childhood to adulthood. *Journal of Experimental Child Psychology*, 114(3), 389–404.
- Rangvid, J., Schmeling, M., & Schrimpf, A. (2013). What do professional forecasters' stock market expectations tell us about herding, information extraction and beauty contests? *Journal of Empirical Finance*, 20, 109–129.
- Rapoport, A., & Chammah, A. M. (1966). The game of chicken. *American Behavioral Scientist*, 10(3), 10–28.
- Repacholi, B., & Slaughter, V. (Eds.). (2004). *Individual differences in theory of mind: Implications for typical and atypical development*. Psychology Press.
- Russell, J. (2013). *Agency: Its role in mental development* [Originally published 1996 by Lawrence Erlbaum Associates]. Psychology Press.
- Selten, R. (1983). Evolutionary stability in extensive two-person games. *Mathematical Social Sciences*, 5(3), 269–363.
- Sher, I., Koenig, M., & Rustichini, A. (2014). Children's strategic theory of mind. *Proceedings of the National Academy of Sciences*, 111(37), 13307–13312.
- Smith, J. M. (1974). The theory of games and the evolution of animal conflicts. *Journal of Theoretical Biology*, 47(1), 209–221.
- Steinberg, L. (2005). Cognitive and affective development in adolescence. *Trends in Cognitive Sciences*, 9(2), 69–74.
- Stone, V. E., Baron-Cohen, S., & Knight, R. T. (1998). Frontal lobe contributions to theory of mind. *Journal of Cognitive Neuroscience*, 10(5), 640–656.
- Sutter, M., Zoller, C., & Glätzle-Rützler, D. (2019). Economic behavior of children and adolescents—a first survey of experimental economics results. *European Economic Review*, 111, 98–121.
- Thaler, R. (1980). Toward a positive theory of consumer choice. *Journal of Economic Behavior & Organization*, 1(1), 39–60.
- Thomson, K. S., & Oppenheimer, D. M. (2016). Investigating an alternate form of the cognitive reflection test. *Judgment and Decision Making*, 11(1), 99–113.
- Toplak, M. E., West, R. F., & Stanovich, K. E. (2011). The cognitive reflection test as a predictor of performance on heuristics-and-biases tasks. *Memory & Cognition*, 39(7), 1275–1289.
- Vasco, M., & De Francisco, M. J. V. (2023). Holt-laury as a gumball machine. Available at SSRN 4491386.
- Xu, S., Pan, Y., Qu, Z., Fang, Z., Yang, Z., Yang, F., Wang, F., & Rao, H. (2018). Differential effects of real versus hypothetical monetary reward magnitude on risk-taking behavior and brain activity. *Scientific Reports*, 8(1), 3712.
- Zelazo, P. D., Carter, A., Reznick, J. S., & Frye, D. (1997). Early development of executive function: A problem-solving framework. *Review of General Psychology*, 1(2), 198–226.

A Appendix I - Additional statistical analysis

Figure 4: Strategic performance by *CRT*, *FinAb* and *Del*



Note: Mean strategic performance in *CARDS* (blue) and *PIGGY* (green) as a function of the number of correct answers in *CRT*, *FinAb* and *Del*. The sum of these three components constitutes the synthetic *CogAb* score. Each row corresponds to one of the three cognitive measures. Within each row, the left panel shows the percentage of participants playing *Nash*, *Trembling* and *Rationality*.

Table 6: Distribution of responses in *CARDS* and *PIGGY*

Card / Seconds	1 / 0-6	2 / 7-12	3 / 13-18	4 / 19-24	5 / 25-30	Total
<i>CARDS</i>	158 (5.8%)	374 (13.8%)	731 (26.9%)	740 (27.2%)	715 (26.3%)	2718
<i>PIGGY</i>	167 (25.6%)	111 (17%)	211 (32.3%)	100 (15.3%)	64 (9.8%)	653

Note: In order to display both experimental designs with the same number of columns, *Secondstopulse* were aggregated in groups of 6.

Table 7: Distribution of responses in control variables

Variable	0	1	2	3	4	5	6	7	8	9	Total
<i>CRT</i>	430 (12.8%)	1056 (31.4%)	1646 (48.9%)	232 (6.9%)							3364
<i>FinAb</i>	946 (28.3%)	1357 (40.5%)	927 (27.7%)	118 (3.5%)							3348
<i>Del</i>	153 (4.6%)	516 (15.6%)	1212 (36.5%)	1436 (43.3%)							3317
<i>CogAb</i>	28 (0.9%)	102 (3.1%)	217 (6.6%)	423 (12.9%)	615 (18.7%)	755 (22.9%)	641 (19.5%)	356 (10.8%)	125 (3.8%)	29 (0.9%)	3291
<i>Risky</i>	19 (0.6%)	96 (2.8%)	363 (10.8%)	1073 (31.8%)	1409 (41.8%)	347 (10.3%)	62 (1.8%)				3369
<i>Patience</i>	585 (17.4%)	309 (9.2%)	248 (7.4%)	387 (11.5%)	524 (15.6%)	960 (28.5%)	355 (10.5%)				3368

Note: Columns are enumerated by the number of reflective or correct answers in *CRT* or *FinAb*, comprehension in probability in *Del*, total punctuation in *CogAb*, number of risky choices in *Risky* and number future choices in *Patience*.

Table 8: Determinants of *Nash*, *Trembling* and *Rationality* by *CogAb*

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	CARDS						PIGGY					
	Nash	Nash	Trembling	Trembling	Rationality	Rationality	Nash	Nash	Trembling	Trembling	Rationality	Rationality
<i>CogAb_std - CARDS</i>		0.002 (0.861)		0.026*** (0.008)		0.011** (0.028)						
<i>CogAb_std - PIGGY</i>								0.020** (0.023)		0.022** (0.029)		0.052*** (0.000)
<i>Age</i>	-0.007 (0.290)	-0.006 (0.310)	-0.002 (0.782)	-0.004 (0.579)	-0.001 (0.892)	-0.001 (0.779)	-0.009 (0.191)	-0.011 (0.118)	-0.004 (0.710)	-0.006 (0.539)	0.006 (0.541)	0.001 (0.951)
<i>Female</i>	-0.004 (0.827)	-0.007 (0.690)	0.001 (0.943)	0.005 (0.802)	0.002 (0.864)	0.003 (0.771)	-0.006 (0.704)	-0.002 (0.905)	-0.014 (0.495)	-0.010 (0.643)	0.017 (0.426)	0.028 (0.162)
<i>Repeater</i>	-0.008 (0.876)	-0.009 (0.871)	-0.044 (0.425)	-0.013 (0.824)	-0.022 (0.430)	-0.010 (0.723)	-0.015 (0.373)	0.004 (0.822)	-0.045* (0.068)	-0.025 (0.277)	-0.118*** (0.002)	-0.071** (0.047)
<i>Class Size</i>	-0.007 (0.122)	-0.007 (0.121)	-0.004 (0.439)	-0.004 (0.410)	-0.003 (0.275)	-0.003 (0.260)	-0.002 (0.620)	-0.002 (0.567)	0.001 (0.864)	0.001 (0.909)	0.002 (0.664)	0.002 (0.739)
<i>RiskyS</i>	0.053 (0.309)	0.058 (0.248)	-0.003 (0.960)	0.018 (0.762)	-0.001 (0.966)	0.010 (0.736)	-0.051 (0.298)	-0.046 (0.327)	-0.091* (0.096)	-0.086 (0.106)	-0.044 (0.555)	-0.033 (0.641)
<i>PatienceS</i>	0.024 (0.320)	0.024 (0.354)	0.065** (0.013)	0.054** (0.041)	0.030** (0.021)	0.025* (0.056)	0.021 (0.346)	0.019 (0.381)	0.025 (0.348)	0.022 (0.377)	0.051 (0.147)	0.046 (0.191)
<i>Second Position</i>							-0.009 (0.620)	-0.009 (0.601)	-0.006 (0.753)	-0.007 (0.733)	-0.009 (0.590)	-0.010 (0.529)
<i>Constant</i>	0.548*** (0.000)	0.544*** (0.001)	0.627*** (0.001)	0.651*** (0.000)	0.786*** (0.000)	0.791*** (0.000)	0.286** (0.027)	0.313** (0.018)	0.270* (0.089)	0.298* (0.065)	0.317* (0.091)	0.386** (0.040)
<i>R-squared</i>	0.009	0.010	0.012	0.015	0.010	0.011	0.095	0.104	0.108	0.115	0.036	0.069
<i>N</i>	2,715	2,636	2,715	2,636	2,715	2,636	646	646	646	646	646	646

Robust pval in parentheses *** p<0.01, ** p<0.05, * p<0.1

Note: Estimation of the impact of cognitive abilities in optimizing *CARDS* and *PIGGY* through linear regressions. For all models we regress the effect of different control variables respect to outcomes (*Nash*, *Trembling* and *Rationality*) while we added for a second specification of each model the effect of the age-adjusted cognitive ability index in standard deviation units for each experimental design subsample. Control variables include age of participants and gender (*Female*), the standardized measures of risk attitudes (*RiskyS*) and patience (*PatienceS*), if

they had to enroll at least twice in the same school year because they failed (*Repeater*) and the class size. Also, to control potential order effects in *PIGGY*, we include the variable *SecondPosition*, which indicates whether *PIGGY* design question was presented in second position of the design. This variable is included only in the regressions related to *PIGGY*. All specifications also include school fixed effects, and all regressions are estimated using heteroskedasticity-robust standard errors clustered at the class level.

Table 9: Determinants of *Nash*, *Trembling* and *Rationality* in *CARDS* by *CRT*, *FinAb* and *Del*

	<i>CARDS</i>														
	(1) Nash	(2) Nash	(3) Nash	(4) Nash	(5) Nash	(6) Trembling	(7) Trembling	(8) Trembling	(9) Trembling	(10) Trembling	(11) Rationality	(12) Rationality	(13) Rationality	(14) Rationality	(15) Rationality
<i>CRT</i>		0.019* (0.095)			0.021* (0.068)		0.036*** (0.004)			0.030** (0.026)		0.017*** (0.007)			0.015** (0.020)
<i>FinAb</i>			0.005 (0.645)		0.002 (0.884)			0.025** (0.034)		0.016 (0.194)			0.010* (0.066)		0.006 (0.281)
<i>Del</i>				-0.015 (0.133)	-0.017* (0.092)				0.008 (0.473)	0.002 (0.891)				0.001 (0.838)	-0.002 (0.736)
<i>Age</i>	-0.007 (0.290)	-0.008 (0.195)	-0.008 (0.255)	-0.005 (0.413)	-0.007 (0.278)	-0.002 (0.782)	-0.005 (0.476)	-0.006 (0.398)	-0.003 (0.680)	-0.008 (0.262)	-0.001 (0.892)	-0.002 (0.584)	-0.002 (0.550)	-0.001 (0.890)	-0.003 (0.463)
<i>Female</i>	-0.004 (0.827)	-0.005 (0.802)	-0.005 (0.786)	-0.007 (0.705)	-0.008 (0.657)	0.001 (0.943)	0.002 (0.932)	0.008 (0.699)	-0.000 (0.993)	0.005 (0.825)	0.002 (0.864)	0.001 (0.887)	0.004 (0.661)	0.001 (0.926)	0.002 (0.805)
<i>Repeater</i>	-0.008 (0.876)	0.004 (0.940)	-0.012 (0.831)	-0.010 (0.856)	-0.006 (0.914)	-0.044 (0.425)	-0.020 (0.722)	-0.029 (0.620)	-0.033 (0.548)	-0.010 (0.856)	-0.022 (0.430)	-0.010 (0.738)	-0.015 (0.600)	-0.019 (0.485)	-0.009 (0.756)
<i>Class size</i>	-0.007 (0.122)	-0.007 (0.114)	-0.007 (0.113)	-0.007 (0.152)	-0.007 (0.130)	-0.004 (0.439)	-0.004 (0.406)	-0.003 (0.485)	-0.004 (0.404)	-0.004 (0.420)	-0.003 (0.275)	-0.003 (0.250)	-0.003 (0.291)	-0.003 (0.275)	-0.003 (0.269)
<i>RiskyS</i>	0.053 (0.309)	0.058 (0.262)	0.065 (0.207)	0.049 (0.332)	0.061 (0.221)	-0.003 (0.960)	0.006 (0.922)	0.011 (0.855)	0.003 (0.959)	0.020 (0.731)	-0.001 (0.966)	0.003 (0.917)	0.006 (0.844)	0.003 (0.930)	0.011 (0.698)
<i>PatienceS</i>	0.024 (0.320)	0.019 (0.448)	0.026 (0.303)	0.026 (0.295)	0.022 (0.407)	0.065** (0.013)	0.054** (0.039)	0.057** (0.028)	0.065** (0.013)	0.052** (0.050)	0.030** (0.021)	0.025* (0.059)	0.027** (0.035)	0.031** (0.020)	0.024* (0.070)
<i>Constant</i>	0.548*** (0.000)	0.540*** (0.001)	0.549*** (0.000)	0.555*** (0.001)	0.555*** (0.001)	0.627*** (0.001)	0.618*** (0.001)	0.629*** (0.001)	0.629*** (0.001)	0.632*** (0.001)	0.786*** (0.000)	0.782*** (0.000)	0.789*** (0.000)	0.782*** (0.000)	0.784*** (0.000)
R-squared	0.009	0.010	0.010	0.010	0.012	0.012	0.015	0.014	0.012	0.016	0.010	0.013	0.011	0.009	0.013
N	2,715	2,709	2,693	2,661	2,636	2,715	2,709	2,693	2,661	2,636	2,715	2,709	2,693	2,661	2,636

Robust pval in parentheses *** p<0.01, ** p<0.05, * p<0.1

Note: Estimation of the impact of *CRT*, *FinAb* and *Del* in optimizing *CARDS* through linear regressions. For all models we regress the effect of different control variables respect to outcomes (*Nash*, *Trembling* and *Rationality*) while we added for a second, third and fourth specification of each model the effect of each variable that composes *CogAb*. In fifth specification we added all these three components. Control variables include age of participants and gender (*Female*), the standardized measures of risk attitudes (*RiskyS*) and patience (*PatienceS*), if they had to enroll at least twice in the same school year because they failed (*Repeater*) and the class size. All specifications also include school fixed effects, and all regressions are estimated using heteroskedasticity-robust standard errors clustered at the class level.

Table 10: Determinants of *Nash*, *Trembling* and *Rationality* in *PIGGY* by *CRT*, *FinAb* and *Del*

	<i>PIGGY</i>														
	(1) Nash	(2) Nash	(3) Nash	(4) Nash	(5) Nash	(6) Trembling	(7) Trembling	(8) Trembling	(9) Trembling	(10) Trembling	(11) Rationality	(12) Rationality	(13) Rationality	(14) Rationality	(15) Rationality
<i>CRT</i>		0.020* (0.059)			0.015 (0.149)		0.020* (0.081)			0.014 (0.241)		0.041*** (0.003)			0.028** (0.039)
<i>FinAb</i>			0.016* (0.080)		0.010 (0.235)			0.014 (0.248)		0.008 (0.536)			0.015 (0.201)		-0.003 (0.829)
<i>Del</i>				0.014* (0.054)	0.009 (0.162)				0.019** (0.032)	0.015* (0.082)				0.067*** (0.000)	0.062*** (0.000)
<i>Age</i>	-0.009 (0.191)	-0.011 (0.137)	-0.012 (0.105)	-0.010 (0.163)	-0.013* (0.093)	-0.004 (0.710)	-0.005 (0.599)	-0.006 (0.510)	-0.005 (0.642)	-0.007 (0.465)	0.006 (0.541)	0.002 (0.775)	0.003 (0.746)	0.003 (0.731)	0.001 (0.872)
<i>Female</i>	-0.006 (0.704)	-0.004 (0.832)	-0.003 (0.879)	-0.006 (0.704)	-0.002 (0.912)	-0.014 (0.495)	-0.012 (0.584)	-0.011 (0.601)	-0.014 (0.494)	-0.010 (0.620)	0.017 (0.426)	0.022 (0.259)	0.020 (0.331)	0.016 (0.414)	0.020 (0.311)
<i>Repeater</i>	-0.015 (0.373)	-0.003 (0.849)	-0.005 (0.763)	-0.009 (0.586)	0.004 (0.816)	-0.045* (0.068)	-0.033 (0.173)	-0.036 (0.135)	-0.036 (0.108)	-0.025 (0.271)	-0.118*** (0.002)	-0.094** (0.011)	-0.109*** (0.004)	-0.090** (0.013)	-0.077** (0.026)
<i>Class size</i>	-0.002 (0.620)	-0.002 (0.506)	-0.002 (0.626)	-0.002 (0.636)	-0.002 (0.548)	0.001 (0.864)	0.000 (0.966)	0.001 (0.860)	0.001 (0.836)	0.001 (0.913)	0.002 (0.664)	0.001 (0.841)	0.002 (0.661)	0.003 (0.557)	0.002 (0.687)
<i>RiskyS</i>	-0.051 (0.298)	-0.047 (0.312)	-0.047 (0.328)	-0.051 (0.296)	-0.046 (0.328)	-0.091* (0.096)	-0.088 (0.101)	-0.087 (0.106)	-0.091* (0.096)	-0.087 (0.105)	-0.044 (0.555)	-0.037 (0.595)	-0.040 (0.584)	-0.046 (0.547)	-0.041 (0.567)
<i>PatienceS</i>	0.021 (0.346)	0.019 (0.410)	0.021 (0.351)	0.021 (0.341)	0.019 (0.395)	0.025 (0.348)	0.022 (0.397)	0.024 (0.353)	0.024 (0.338)	0.022 (0.378)	0.051 (0.147)	0.046 (0.212)	0.051 (0.150)	0.051 (0.134)	0.047 (0.181)
<i>Second position</i>	-0.009 (0.620)	-0.008 (0.641)	-0.010 (0.572)	-0.008 (0.624)	-0.009 (0.603)	-0.006 (0.753)	-0.006 (0.773)	-0.007 (0.710)	-0.006 (0.757)	-0.006 (0.744)	-0.009 (0.590)	-0.008 (0.630)	-0.010 (0.541)	-0.008 (0.602)	-0.008 (0.632)
<i>Constant</i>	0.286** (0.027)	0.290** (0.025)	0.302** (0.023)	0.258** (0.040)	0.280** (0.030)	0.270* (0.089)	0.273* (0.085)	0.284* (0.082)	0.231 (0.131)	0.250 (0.111)	0.317* (0.091)	0.325* (0.073)	0.333* (0.086)	0.181 (0.304)	0.195 (0.262)
R-squared	0.095	0.101	0.099	0.098	0.104	0.108	0.112	0.110	0.112	0.115	0.036	0.051	0.038	0.078	0.084
N	646	646	646	646	646	646	646	646	646	646	646	646	646	646	646

Robust pval in parentheses *** p<0.01, ** p<0.05, * p<0.1

Note: Estimation of the impact of *CRT*, *FinAb* and *Del* in optimizing *CARDS* through linear regressions. For all models we regress the effect of different control variables respect to outcomes (*Nash*, *Trembling* and *Rationality*) while we added for a second, third and fourth specification of each model the effect of each variable that composes *CogAb*. In fifth specification we added all these three components. Control variables include age of participants and gender (*Female*), the standardized measures of risk attitudes (*RiskyS*) and patience (*PatienceS*), the class size and school type (*Public*) and the device used to complete the survey (*Phone* and *Tablet*). Also, to control potential order effects in *PIGGY*, we include the variable *SecondPosition*, which indicates whether *PIGGY* design question was presented in second position of the design. All specifications also include school fixed effects, and all regressions are estimated using heteroskedasticity-robust standard errors clustered at the class level.

Table 11: Determinants of *Secondstopulse* in *PIGGY*

	(1) Seconds no FE	(2) Seconds FE School	(3) Seconds FE Class	(4) Seconds no FE	(5) Seconds FE School	(6) Seconds FE Class	(7) Seconds no FE	(8) Seconds FE School	(9) Seconds FE Class
<i>Computer</i>	-1.908 (0.581)	-2.024 (0.586)	-1.076 (0.722)						
<i>Tablet</i>				1.845* (0.084)	1.630 (0.125)	1.518 (0.185)			
<i>Phone</i>							-1.729 (0.100)	-1.518 (0.150)	-1.446 (0.196)
<i>Constant</i>	13.658*** (0.000)	13.345*** (0.000)	7.083*** (0.000)	13.290*** (0.000)	12.915*** (0.000)	6.957*** (0.000)	15.031*** (0.000)	14.427*** (0.000)	8.408*** (0.000)
R-squared	0.000	0.007	0.079	0.008	0.013	0.083	0.007	0.012	0.083
<i>N</i>	653	653	653	653	653	653	653	653	653

Robust pval in parentheses *** p<0.01, ** p<0.05, * p<0.1

Note: Estimation of the impact of time of response in *PIGGY* through linear regressions. For each models we regress the effect of *Secondstopulse* for each device including no fixed effects (models 1, 4 and 7), school fixed effects (models 2, 5 and 8) and class fixed effects (models 3, 6 and 9). All specifications are estimated using heteroskedasticity-robust standard errors clustered at the class level.

B Appendix II - Supplementary Materials

B.1 Games' Screenshots

Box 1 - *CARDS* experimental design

Screen 1:

You are now going to be **paired** with another participant, but you will **NOT know** who they are.

Both of you see the same cards. And each of you **choose one**, at the same time and without speaking.

- If both of you choose the **same card**, YOU get **the number of points shown on that card** and THE OTHER PERSON gets **0 points**.
- If you do **NOT** choose the **same card**, EACH OF YOU gets **the number of points shown on your chosen card**.



Which card do you choose?

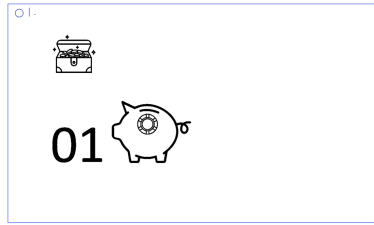
- 1
- 2
- 3
- 4
- 5

Box 2 - *PIGGY* experimental design

Screen 1:

You are now going to be paired with another participant, but you will not know who they are. The other person will not know who you are either.

Imagine a **piggy bank** like the one below that receives one euro every second for **30 seconds**. For example: after one second it will contain €1, after two seconds €2, and so on until it reaches €30, after which it will be destroyed.



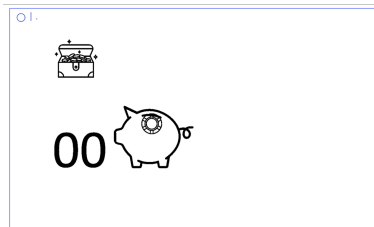
Screen 2:

In this case, the money will go to the person who presses the piggy bank **SECONDLY**, keeping the **euros** it contains at that moment. You can press anywhere on the piggy bank to stop it. Remember:

- If you are the second participant to press, you will earn the number of euros in the piggy bank at that moment.
- If you are the first participant to press, or if you do not press within 30 seconds, you earn 0 euros.
- If both of you press at the same time, it's a tie and the euros in the piggy bank will be split equally between you.

Screen 3:

The piggy bank goes to the **SECONDLY** person to press it, with the euros it contains at that moment.



B.2 *CognitiveReflectionTest(CRT)*

The items used in the *CRT* task are shown in Figure 5. Reflective and intuitive answers are shown in parentheses. The order of the items was randomized.

Figure 5: *CRT* questions

Q1. Emilia’s father has three daughters. The first two are named April and May. What is the name of the third daughter?

(**Reflective: Emilia**; Intuitive: June)

Q2. If you are running a race and you overtake the person in second place, what position are you in? Indicate with a number. For example: 1 (first), 2 (second), etc.

(**Reflective: 2**; Intuitive: 1)

Q3. In a library, the number of books doubles every month. If it takes 48 months for the library to become full, how many months would it take for it to be half full? Indicate with a number.

(**Reflective: 47**; Intuitive: 24)

B.3 *FinancialAbilities*

The items used in the *FinAb* task are shown in Figure 6. Correct answers are shown in parentheses or in bold. The order of the items was randomized.

Figure 6: *FinAb* questions

Q1. If there are 5 people who own the winning lottery ticket and the prize to be shared is 2 million euros, how much money will each person receive? Write the amount using numbers only, without dots or commas.

(**Correct answer: 400,000**)

Q2. Imagine you have €100 in a savings account. The account earns an interest rate of 10% per year. How much money will you have in the account after 2 years?

(**Correct answer: 121**)

Q3. Imagine you have €100 in a savings account and the interest rate you earn is 2% per year. If you keep the money in the account for 5 years, how much money will you have at the end of those 5 years?

- Less than 102€
- 102€ exactly
- **More than 102€**
- I don’t know

B.4 *DelavandeTest*

A violation of the Delavande test occurs when:

- The value reported for the first question is lower than that reported for the second.
- The value reported for the third question is lower than that reported for the fourth.
- The value reported for the fifth question is lower than that reported for the sixth.

Note that participants in the first 10 schools responded to the task by entering a number, while in the rest 2 schools of the sample a half of participants entered numbers and the other half moved a slider with randomization at the individual level (see Figure 7). The instructions and questions were maintained in both versions.

Figure 7: Slider

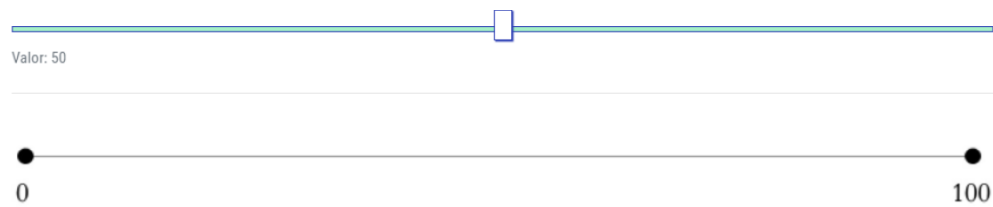


Figure 8: Screen 1

In the next screen, we **will ask you** about the **probability** that certain things could happen. In each question, you should **write down** how **likely you think** it is that these things will happen in reality.

If you think something will happen for **sure**, you should write **100**. If you are fairly but not completely sure, write a number close to 100 (but less than 100). If you think it is something that will **never** happen, write **0**. If you think it is unlikely, write a number close to 0 (but greater than 0). If you think it is **equally likely** to happen or not to happen, write **50**.

Figure 9: Screen 2

EXAMPLES:

- If it is a sunny day and I ask you how likely it is that it will be sunny within 1 hour, you would typically say 100 or numbers very close to 100.
- If it is 12 o'clock at night, the most reasonable thing to say is that the probability that it will be sunny in 1 hour is 0.
- If I flip a coin, the probability that it will come up heads is 50.

Figure 10: *Del* questions

Q1. Imagine that I have a basket with 5 apples: 1 green and 4 red. If I ask you to choose one of the apples without looking inside the basket, how likely is it that you pick the green apple, from 0 to 100?

Q2. Imagine that I have a basket with 10 apples: 1 green and 9 red. If I ask you to choose one of the apples without looking inside the basket, how likely is it that you pick the green apple, from 0 to 100?

Q3. How likely is it that you eat rice in the next week from 0 to 100? (Including today)

Q4. How likely is it that you eat rice in the next month from 0 to 100? (Including today)

Q5. How likely is it that you do not attend your educational center even once in the next month from 0 to 100? (Including today)

Q6. How likely is it that you take a shower at least once during the next month from 0 to 100? (Including today)

B.5 *TheTrukTask*

Figure 11: Screen 1

Throughout the next 6 screens, you will have to make 6 decisions (one per screen) about **how you want to receive a hypothetical amount of money**. Your task is to choose which option you prefer, knowing that if you choose one you will receive the money **tomorrow**, and if you choose the other you will receive it **next week**. As you progress through the decisions, the amount of money you will receive for waiting will get **larger and larger**.

Figure 12: Screen 2

¿Qué prefieres?

☐ A)



☐ B)



Figure 13: Screen 3

¿Qué prefieres?

☐ A)



☐ B)



Figure 14: Screen 4

¿Qué prefieres?

☐ A)



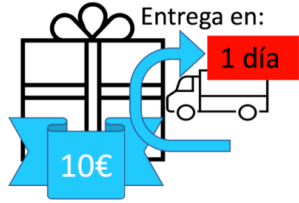
☐ B)



Figure 15: Screen 5

¿Qué prefieres?

☐ A)



☐ B)

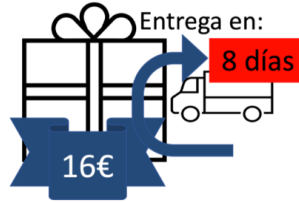
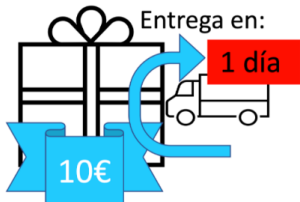


Figure 16: Screen 6

¿Qué prefieres?

☐ A)



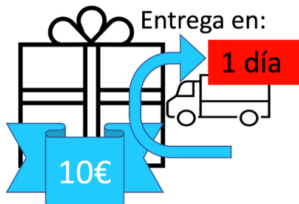
☐ B)



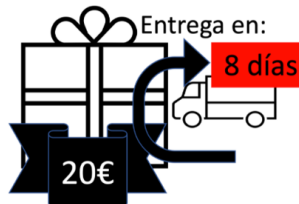
Figure 17: Screen 7

¿Qué prefieres?

☐ A)



☐ B)



B.6 *TheGumballMachine*

Figure 18: Screen 1

In this task you are going to make decisions about **probabilities**. Probability is the possibility of something happening, that is, how someone is sure what is going to happen.

In each decision **you will have to choose between two options** where you can win fictitious money: in both you will win, but you can be luckier and win more, or you can be less lucky and win less. **The probabilities of winning change** from one decision to another.

Below, you are presented with **6 different questions**. Your task is to choose option **A)** or option **B)** in all questions.

To complete the task, proceed to the **next screen**.

Figure 19: Screen 2

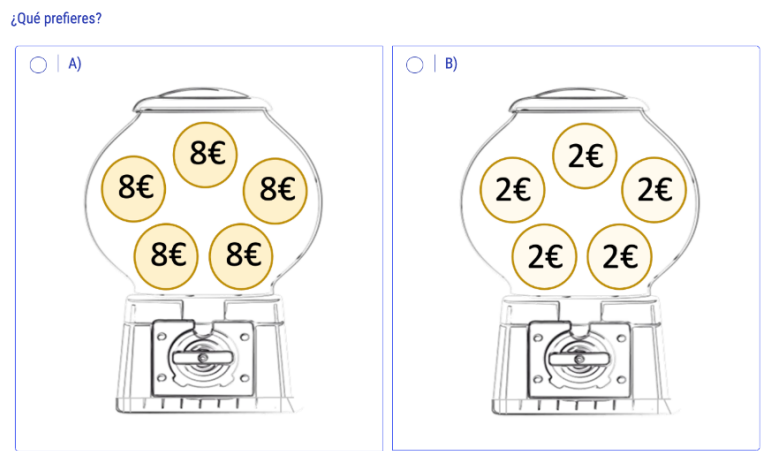


Figure 20: Screen 3

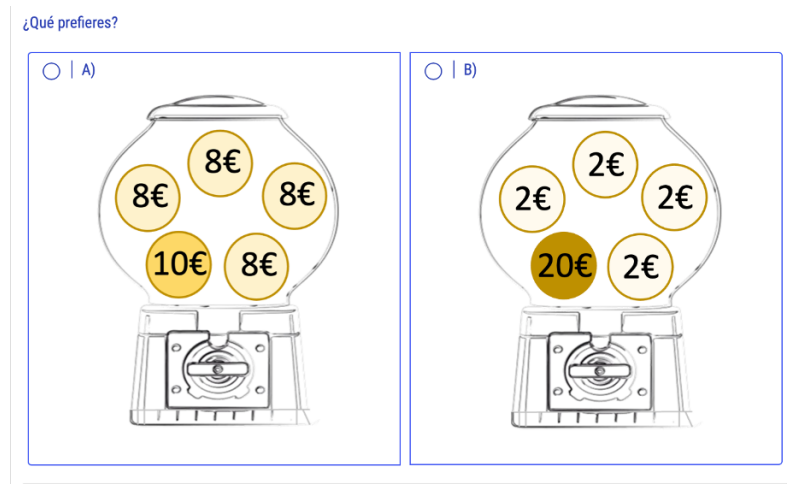


Figure 21: Screen 4

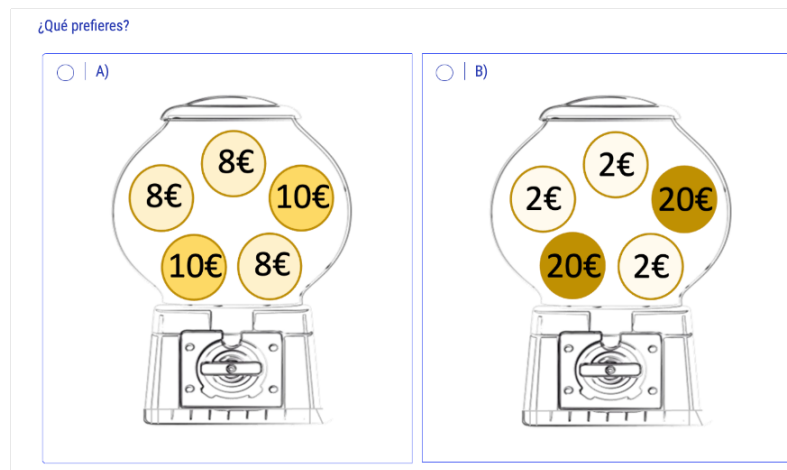


Figure 22: Screen 5

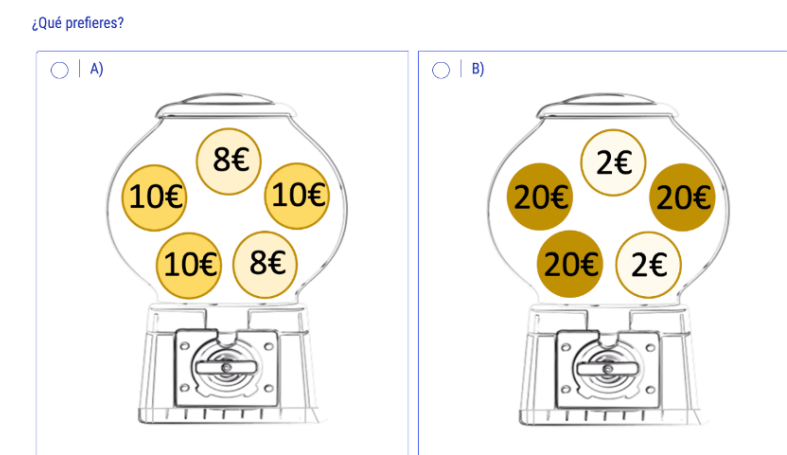


Figure 23: Screen 6

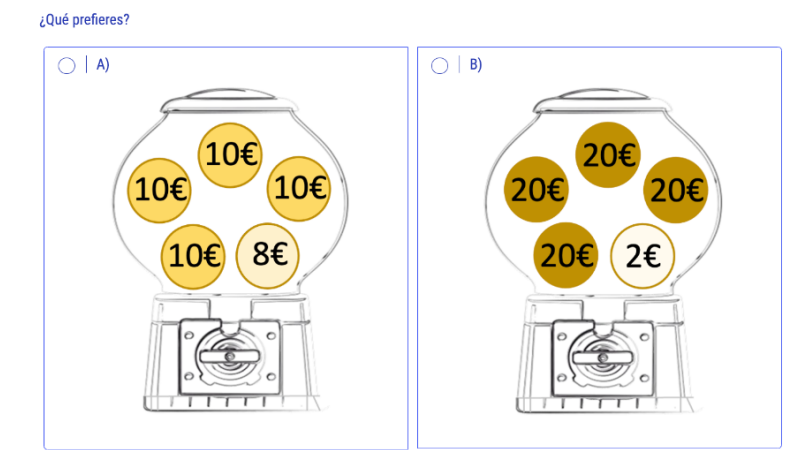


Figure 24: Screen 7

